

StellarOne

Installation Guide

Version 5.0
Patch 2



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<https://my.txone.com/>

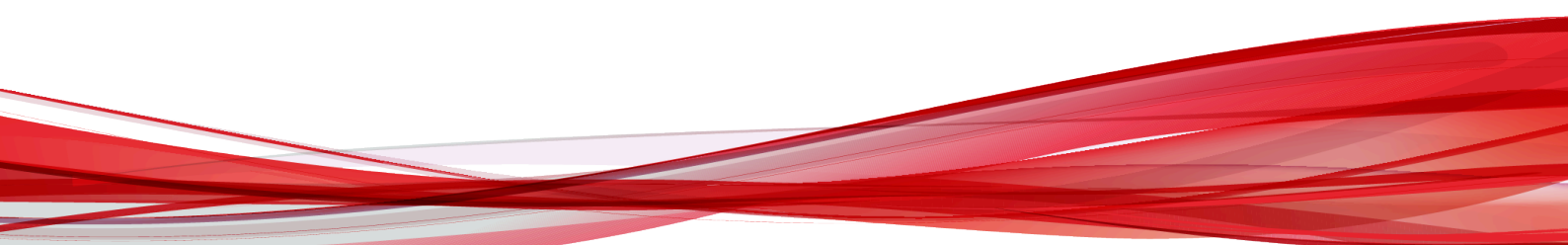
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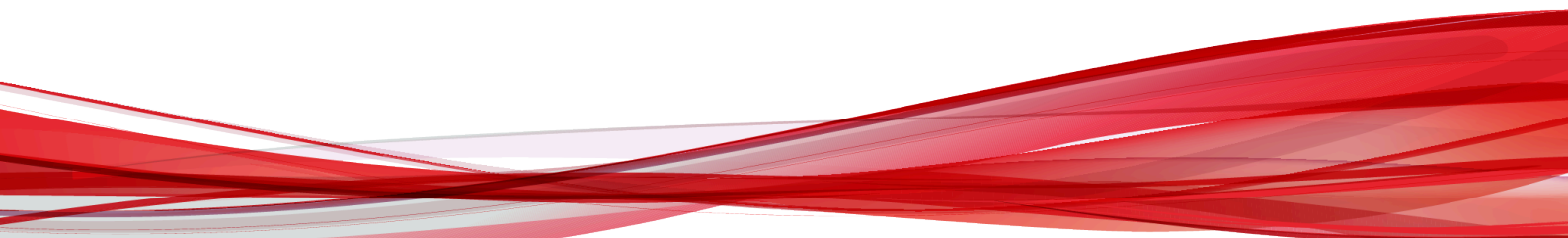


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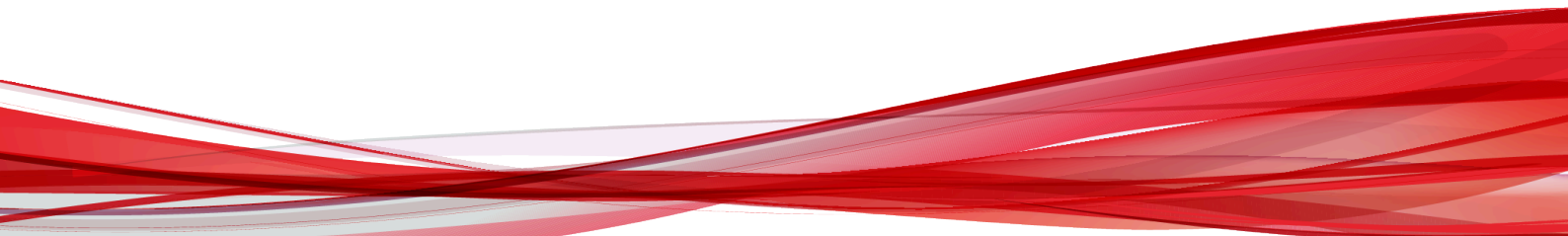
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Preface

This guide introduces TXOne StellarOne™ and covers all aspects of product management.

Topics in this chapter include:

- [About the Documentation \(on page vii\)](#)
- [Audience \(on page vii\)](#)
- [Document Conventions \(on page vii\)](#)
- [Terminology \(on page viii\)](#)



About the Documentation

TXOne StellarOne™ documentation includes the following:

Documentation	Description
Readme file	Contains a list of known issues and basic installation steps. It may also contain late-breaking product information not found in the other documents.
Installation Guide	A PDF document that discusses requirements and procedures for installing StellarOne
Administrator Guide	A PDF document that discusses StellarOne agent installation, getting started information, and server and agent management
Online Help	HTML files that provide "how to's", usage advice, and field-specific information
Knowledge Base	An online database of problem-solving and troubleshooting information. It provides the latest information about known product issues. To access the Knowledge Base, go to the following websites: https://kb.txone.com/

Audience

TXOne StellarOne™ documentation is intended for administrators responsible for StellarOne management, including agent installation. These users are expected to have advanced networking and server management knowledge.

Document Conventions

The documentation uses the following conventions.

Table 1. Document Conventions

Convention	Description
UPPER CASE	Acronyms, abbreviations, and names of certain commands and keys on the keyboard
Bold	Menus and menu commands, command buttons, tabs, and options

Table 1. Document Conventions (continued)

Convention	Description
<i>Italics</i>	References to other documents
<code>Monospace</code>	Sample command lines, program code, web URLs, file names, and program output
Navigation > Path	The navigation path to reach a particular screen For example, File > Save means, click File and then click Save on the interface
 Note:	Configuration notes
 Tip:	Recommendations or suggestions
 Important:	Information regarding required or default configuration settings and product limitations
 Warning:	Critical actions and configuration options

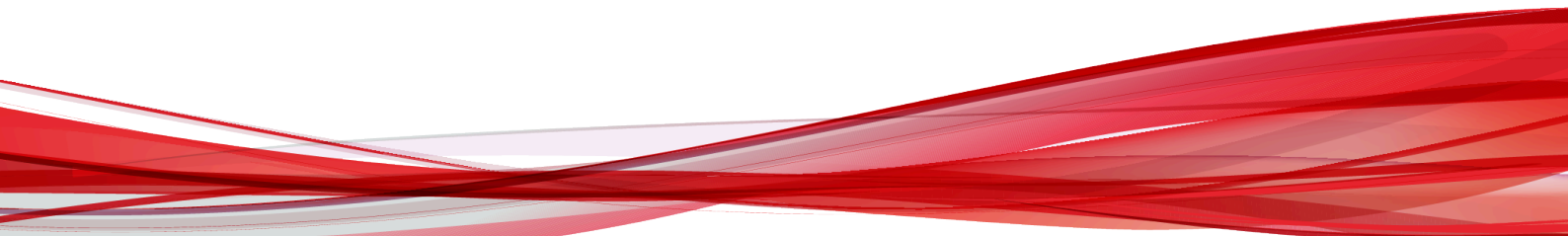
Terminology

The following table provides the official terminology used throughout the TXOne StellarOne™ documentation:

Terminology	Description
server	The StellarOne console server program
server endpoint	The host where the StellarOne server is installed

Terminology	Description
agents	The host running the Stellar Agent program
managed agents managed endpoints	The hosts running the Stellar Agent program that are known to the StellarOne server program
target endpoints	The hosts where the StellarOne managed agents will be installed
Administrator (or StellarOne administrator)	The person managing the StellarOne server
StellarOne (management) console	The user interface for configuring and managing StellarOne settings and the agents managed by StellarOne
CLI	Command Line Interface
license activation	Includes the type of StellarOne server installation and the allowed period of usage that you can use the application
primary license	The license that determines the StellarOne server edition. When a Discover add-on license is in use, the primary license is always a StellarICS license.
add-on license	An additional license entitlement applied alongside the primary license. An add-on license provides a separate seat pool but does not change the server edition.
Discover agent	A managed agent running Discover Edition. Discover agents provide visibility only and do not perform real-time protection.
protection agent	A managed agent that is not running Discover Edition, such as an agent running Protect Edition.
hard expiration date	The expiration date of a license plus its grace period. A license or add-on license remains effective until its hard expiration date is reached.

Terminology	Description
agent installation folder	The folder on the host that contains the Stellar Agent agent files. If you accept the default settings during installation, you will find the installation folder at one of the following locations: C:\Program Files\TX0ne\StellarProtect C:\Program Files\TX0ne\StellarProtect (Legacy Mode)



Chapter 1. Introduction

This section introduces TXOne StellarOne™ and provides an overview of its features.

Topics in this chapter include:

- [About TXOne Stellar \(on page 12\)](#)
- [Key Features and Benefits \(on page 13\)](#)
- [What's New \(on page 14\)](#)

About TXOne Stellar

TXOne Stellar provides a context-focused security solution for OT endpoints and cyber-physical systems (CPS), aiming to defend operation stability with continuous detection and response aligned to the specific requirements of the OT domain.

TXOne Stellar platform is composed of the centralized management console server and unified agents apt for legacy OT devices and modern cyber-physical systems.

- StellarOne™, designed to streamline administration of the agents installed on modernized systems and legacy systems, along with its intuitive centralized management, consistent policy enforcement, and action-oriented alerts that empower security teams of all sizes and skill levels to successfully mature their organization's security posture.
- Stellar Agent™ / Stellar Agent (Windows-Legacy) / Stellar Agent (Linux), using a single-agent design that delivers seamless asset-centric protection and ensures coverage for modern CPS and legacy OT devices throughout their entire asset lifecycle. The lightweight unified agent simplifies security by combining CPS Detection and Response (CPSDR), threat prevention, operations lockdown, device control.
 - CPSDR: Embodied within the advanced Operations Behavior Anomaly Detection feature, which establishes a unique baseline of each agent-device during practicable operating states and performs deviation analysis by means of an expansive industrial application repository and ransomware detection engine to defend against unexpected changes that may impact stability.

Moreover, TXOne Stellar brings the contextualization of security into an operation-led view to allow both the operation and security teams to achieve their goals without needing to compromise. To illustrate, if a device suddenly tried to start launching different applications, it would be blocked from doing so.

From the operation view, this may be an unplanned auto-update that, if run, would take the device offline to reboot. From a security view, this could be an attempt to access an encryption library that is about to be used to execute ransomware. By applying the operation context, both security and operation-initiated changes can be detected, and appropriate responses are taken.

In both cases, CPSDR stopped the incident before it could occur. The security team followed up and resolved the ransomware infection in a different part of the environment. The operation team could schedule the required update to run during planned maintenance period.

- Multi-Method Threat Prevention: Provides advanced threat scan on the basis of ICS root of trust and operations-focused machine learning to secure the agent-devices against known and unknown malware threats without compromising operational availability.
- Operations Lockdown: For fixed-function and devices with limited patching availability, operations lockdown enforcement prohibits unauthorized changes, including alterations to registry and function parameters.
- Trusted Peripheral Control: Unauthorized access from external sources, such as USB devices, is configurable and controlled to reduce physical access threats.

Leveraging an expansive OT application and certificate library and exclusive ransomware detection engine, TXOne Stellar maintains CPS operational integrity through behavioral anomaly detection and eliminates configuration drift for legacy and fixed-use assets with device lockdown. Security teams can confidently deliver detection and response outcomes across the OT terrain, with TXOne Stellar effectively strengthen organization's security posture while maintaining its business operations stability.

Key Features and Benefits

The TXOne StellarOne™ management console provides following features and benefits.

Table 2. Features and Benefits

Feature	Benefit
VDS (Vulnerability Detection Service)	VDS (Vulnerability Detection Service) is a security visibility feature that enables Stellar agents, including the Discover Edition, to automatically collect endpoint and asset information and report it to the StellarOne console. The console then cross-references this data against a regularly updated vulnerability pattern database to identify known vulnerabilities (CVEs) affecting those endpoints. Each detected vulnerability is also assigned a severity level so that administrators can quickly identify high-risk assets and prioritize remediation efforts. Vulnerability data is surfaced through the StellarOne console and dashboard, providing centralized visibility into the organization's security posture — whether used alongside full protection agents or as a standalone assessment tool.
Cyber-Physical System Detection and Response (CPSDR)	The CPSDR requires a deep understanding of what the expected behaviors for each device are. Embodied within the advanced Operations Behavior Anomaly Detection feature, which primarily defends against unexpected changes that may impact operational stability by comparing daily operation processes and behaviors with a unique baseline of each agent-device and performing comprehensive behavioral analysis not only via identifying baseline deviation but also using TXOne Networks' exclusive industrial application repository and ransomware detection engine.
Dashboard	The Dashboard provides an overview of the CPS situational awareness and agent deployment status at the centralized control level, along with the monitored asset status and StellarOne console's system status. Administrators can check deployed agent status easily, have quick access to the detected events for further investigation or actions if needed, and generate security reports related to specific agent activity for specified periods.
Centralized Agent Management	StellarOne allows administrators to perform the following tasks:

Table 2. Features and Benefits (continued)

Feature	Benefit
	• Monitor Stellar Agent (Windows)/Stellar Agent (Windows-Legacy) agent status • Examine connection status • View configurations • Collect agent logs on-demand • Turn agent Application Lockdown on or off • Enable or disable agent Device Control • Configure agent Maintenance Mode settings • Update agent components • Initialize the Approved List • Deploy agent patches • Add trusted files and USB devices • Export agents information • Import/Export agents configuration settings and import agent Approved list • Export agents Approved List
Centralized Event Management	On endpoints protected by Stellar Agent (Windows)/Stellar Agent (Windows-Legacy) agents, administrators can monitor status and events, as well as respond when files are blocked from running. StellarOne provides event management features that let administrators quickly know about and take action on the blocked-file events.
Server Event Auditing	Operations performed by StellarOne web console accounts are logged. StellarOne records an operating log for each account, tracking who logs on, who deletes event logs, and more.

What's New

TXOne StellarOne™ 5.0 provides following new features and enhancements.

Table 3. What's New in TXOne StellarOne™ 5.0

Feature	Description
Discover add-on license	A StellarDiscover license can be applied as an add-on to an existing StellarICS primary license, giving Discover agents a dedicated seat pool so they do not consume StellarICS seats.
Discover Edition	Support for the new visibility-focused Discover Edition of Stellar Agent
VDS integration visibility	New security monitoring widgets leveraging VDS (vulnerability/detection source)
New dashboard widgets for asset management	Additional widgets on the StellarOne dashboard focused on asset-centric views
New widget: Threat type count per agent	Dashboard widget that aggregates threat detections by type per agent
New remote emergency stop feature	Ability to remotely emergency stop protection from StellarOne
Random distribution of Agent Scan Component Update Schedule	Randomly distribute agent component update schedule across an hour. This reduces server load, allowing more agents to update.

Chapter 2. Installation Planning

This chapter shows how to plan for TXOne StellarOne installation.

Topics in this chapter include:

- [System Requirements \(on page 18\)](#)
- [Hardware Requirements \(on page 18\)](#)
- [Planning Network Bandwidth for Agent Deployment \(on page 21\)](#)
- [Outbound Ports and FQDN Used \(on page 23\)](#)
- [Supported Agent Versions \(on page 26\)](#)
- [Instance Data Encryption \(on page 29\)](#)

System Requirements

TXOne StellarOne™ is packaged in an Open Virtual Appliance (OVA), Virtual Hard Disk (VHD), Virtual Hard Disk v2 (VHDX) and QEMU Copy On Write Version 2 (QCOW2) formats. The above-mentioned package files respectively apply to different hypervisors.

Supported Hypervisors (OVA file)

- VMware ESXi 6.5 or above
- VMware Workstation 16 Pro or above

Supported Hypervisors (VHDX file)

- Windows Server 2019, Hyper-V Manager Windows 10 or above

Supported Hypervisors (QCOW2 file)

- QEMU v1.7 or above



Note:

For StellarOne deployed from AMI on a AWS EC2 instance, see [Hardware Requirements for Deploying StellarOne on VMware/Hyper-V/AWS EC2 \(on page 19\)](#) for more details.

Supported Browser

- Google Chrome 87 or above
- Microsoft Edge 79 or above
- Mozilla Firefox 78 or above

Supported Cloud Platforms (VHD file)

- Microsoft Azure Virtual Machine

Minimum Supported Resolution

- 1366x768

Hardware Requirements

Hardware requirements vary depending on the number of agents that will be configured and retained, as well as features enabled. See the sections below for determining the optimal number of agents that your StellarOne server deployment can manage on different platforms.

Hardware Requirements for Deploying StellarOne on VMware/Hyper-V/AWS EC2

See the following table for determining the optimal number of agents that your StellarOne instance can manage on the VMware, Hyper-V, AWS EC2 and Azure VM platform.



Note:

To deploy StellarOne on AWS EC2, please also refer to [Amazon EC2 Instance Types](#) for specifications of the instance types.

Table 4. Sizing Table for Deploying StellarOne

Max. No. of Agents	Min No. of vCores	Memory Size	1st HDD Space	2nd HDD Space (Minimum)	2nd HDD Space (Recommended)
30,000	10	32 GB	45 GB	120 GB	520 GB
20,000	8	16 GB		120 GB	420 GB
15,000	8	16 GB		70 GB	320 GB
10,000	8	16 GB		70 GB	220 GB
5,000	8	16 GB		70 GB	170 GB
1,000	4	16 GB		70 GB	120 GB
500	4	12 GB		70 GB	120 GB



Important:

StellarOne requires one external disk with at least 70 GB minimum space for initialization and booting process.

- The minimum required 2nd HDD space is used to store the system configurations and event logs.
- The recommended required 2nd HDD space is used to store the system configurations and event logs, as well as executing the **Operations Behavior Anomaly Detection** feature.



For example, if you want to deploy 25,000 agents **without** running the **Operations Behavior Anomaly Detection** feature, you'll need to prepare a 2nd HDD of 120 GB capacity. However, if you want to deploy 25,000 agents and run **Operations Behavior Anomaly Detection**, you'll need to prepare a 2nd HDD of 520 GB capacity

**Note:**

1. You may reuse the external disk of a terminated StellarOne instance if you want to migrate the previous configurations and logs to a new StellarOne instance.
2. Please also take the network bandwidth into consideration when planning for agent deployment. See [Planning Network Bandwidth for Agent Deployment \(on page 21\)](#) for more details.

Version 3.3 or Later Intermediate Upgrade Requirement

Before upgrading to StellarOne version 3.3 or any later version, users must first upgrade to version 3.2.5. Upgrading to version 3.3 from a version earlier than 3.2.5 will result in automation and login errors.

The longer a specific instance of StellarOne has been installed and in use, the memory required to install version 3.3 or any later version increases. The table below shows the minimum memory required based on number of years of StellarOne use.

Table 5. 3.3 Upgrade Sizing Table

Usage Duration (Years)	Minimum Memory (GB)
3 or less	12
4	12
5	12
6	12
7	12
8	13
9	14
10	15



Note: The memory required based on usage duration is an estimation and will vary slightly depending on actual usage. If memory or HDD free space is insufficient, an error message will be displayed. After resolving the issue, re-perform the firmware upgrade.

Please also note that upgrade may fail if there is:

1. Less than 15 GB of free space on the 1st HDD pre-installation
 - a. In this case, please increase the free disk space to 45 GB or larger
2. Less than 5 GB free space on the 2nd HDD's System partition pre-installation
3. Less than 30% free space on the 2nd HDD's Database partition pre-installation

You can free up space on the 2nd HDD by:

1. Deleting unused patch files
2. Purging unwanted agent debug logs

If free disk space on the 2nd HDD is still insufficient, please increase the disk size. When deciding what disk size to upgrade to, you can refer to the below table to better understand how much free space will be left remaining on the 2nd HDD after installation of various versions of StellarOne. For example, installing 3.3, a 70 GB HDD would leave 55 GB free space on the Database partition, and 8 GB free space on the System partition. Different versions of StellarOne will leave different amounts of remaining storage space post installation, for example, on a 70 GB HDD, installing 3.2 would leave the device with 12 GB of free space on the System partition, which is enough.



Note: Although the below table shows that the free space post installation for 3.3 is more than 3.2 Service Pack 1, this is only the case for fresh installations. When upgrading from one version to another, the amount of free space post installation will never be less than the previous version, because the previous version's files will still be on the device. After installing version 3.2 Service Pack 1 on a 70 GB HDD, the System partition will have 12 GB of available space. If you want to then upgrade to 3.3, there will be enough space to do so, since 5 GB is required.

Table 6. 2nd HDD Free Space Post Installation

Installation Version	Partition	70 GB HDD	Upgrade to 80 GB HDD (from 70 GB HDD)	Upgrade to 90 GB HDD (from 70 GB HDD)
3.3	Database	55 GB	+8 GB	+16 GB
	System	8 GB	+2 GB	+4 GB
3.2 Service Pack 1	Database	51 GB	+8 GB	+15 GB
	System	12 GB	+3 GB	+5 GB
3.2	Database	51 GB	+8 GB	+15 GB
	System	12 GB	+3 GB	+5 GB

Planning Network Bandwidth for Agent Deployment

Please take network bandwidth into consideration when planning for agent deployment. See below as an example of calculating the bandwidth required to support the number of agents planned to deploy.

Basic concept:

Total available bandwidth / Deployment task size = How many agents can be deployed at one task

Currently, there are 3 types of StellarOne deployment tasks:

- Incremental Pattern Update: works for agent pattern version no less than server version for two weeks, which requires about less than 5 MB
- Full Pattern Update: works for agent pattern version that's already exceeded two-week duration compared to server/update source, which requires about 80 MB
- Agent Remote Patch: update with the remote agent deployment upgrade package, which requires about 110 MB

The following tables illustrate the number of agents to be deployed on condition that the deployment takes 5 minutes and requires 50% of network bandwidth, as well as the recommended policy refresh interval regarding the number of agents managed.

Table 7. Agent Deployment Plan

Total Bandwidth / Deployment Task	No. of Agents Deployed		
	10 Mbps	100 Mbps	1000 Mbps
Incremental Pattern Update	38	375	1000*
Full Pattern Update	2	23	234
Agent Remote Patch	3	27	268

(*) It is not recommended to update more than 1000 agents at the same time. Triggering updates simultaneously for large number of agents uses up substantial computing resources. Some updates may fail when there are other activities on StellarOne.

To keep large numbers of agents up to date, please make use of "Agent Component Update Schedule" or "Update Agents" in agent policy settings.

Table 8. Policy Refresh Interval vs No. of Agents Managed

Policy Refresh Interval	No. of Agents Managed
5 minutes	5000
10 minutes	10000
20 minutes	20000
60 minutes	30000

Outbound Ports and FQDN Used

The following table shows the ports used by the StellarOne server. Please keep them opened in your firewall settings for StellarOne's use.

Table 9. Outbound Ports and FQDN

From	To	Open Port	FQDN	Function
Stellar Agent	StellarOne	9443, 8000, 443	-	StellarOne's listening port for Stellar Agent
Stellar Agent (Windows-Legacy)	StellarOne	8000, 443	-	StellarOne's listening port for Stellar Agent (Windows-Legacy)
StellarOne	Stellar Agent	14336, 14337	-	Stellar Agent's listening port
StellarOne	Stellar Agent (Windows-Legacy)	14336, 14337	-	Stellar Agent (Windows-Legacy)'s listening port
StellarOne	License (PR) Server	443	https://odc.cs.txone-networks.com https://apis.cs.txone.com	For StellarOne versions 3.2.1 and before. StellarOne connects to global server port for license verification and renewal through HTTPS
StellarOne	License (PR) Server	443	https://apis.cs.txone.com	For StellarOne 3.2.3 and later. StellarOne connects to global server port for license verification and renewal through HTTPS

Table 9. Outbound Ports and FQDN (continued)

From	To	Open Port	FQDN	Function
Browser	StellarOne Web	443	-	StellarOne's listening port for web access through HTTPS
StellarOne	Active Update Server	443	https://txse-p.activeupdate.trendmicro.com/activeupdate https://txsp-p.activeupdate.trendmicro.com/activeupdate https://odc.cs.txone-networks.com	For StellarOne 2.0 and before. StellarOne connects to global server port for the Stellar Active Update through HTTPS
StellarOne	Active Update Server	443	https://ttauc.s.txone.com/protect https://ttauc.s.txone.com/enforce https://odc.cs.txone-networks.com	For StellarOne versions between 2.1 and 3.1.3. StellarOne connects to global server port for the Stellar Active Update through HTTPS
StellarOne	Active Update Server	443	https://ttauc.s.txone.com/protect https://ttauc.s.txone.com/enforce https://apis.cs.txone.com https://storage.cs.txone.com	For StellarOne 3.2.1. StellarOne connects to global server port for the Stellar Active Update through HTTPS

Table 9. Outbound Ports and FQDN (continued)

From	To	Open Port	FQDN	Function
			https://odc.cs.txone-networks.com	
StellarOne	Active Update Server	443	https://ttaucs.txone.com/protect https://ttaucs.txone.com/enforce https://apis.cs.txone.com https://storage.cs.txone.com	For StellarOne 3.2.3 and after. StellarOne connects to global server port for the Stellar Active Update through HTTPS



Note: If your file wall settings support wildcard characters, you can simply add the following URL: https://*.cs.txone.com



Note:

The following ports are reserved for StellarOne private service usage and are not allowed to use for other purposes.

Table 10. StellarOne Occupied Ports

StellarOne Occupied Port	Port
Stellar Agent (Windows-Legacy) Default Port	8000
Stellar Agent (Windows) Default Port	9443, 8000
SSH	22
NTP	123

Table 10. StellarOne Occupied Ports (continued)

StellarOne Occupied Port	Port
Web	443
StellarOne Internal Service	25
	7590
	8888
	8889
	8999
	9091

Supported Agent Versions

The following table indicates the StellarOne supported agent versions.



Warning:

- Before upgrading, please check the table below to identify the StellarOne supported agent versions.
- Please upgrade the StellarOne server first before you upgrade the agents.

Table 11. Supported Agent Version

Server Version	Agents Version			
StellarOne	Stellar Agent (Windows) (Formerly StellarProtect)	Stellar Agent (Windows-Legacy) (Formerly StellarProtect (Legacy Mode))	Stellar Agent Linux (Formerly StellarProtect Linux)	StellarEnforce
5.0	Protect Edition: 5.0 and earlier versions	Protect Edition: 5.0 and earlier versions	1.0 and earlier versions	N/A

Table 11. Supported Agent Version (continued)

Server Version	Agents Version			
StellarOne	Stellar Agent (Windows) (Formerly StellarProtect)	Stellar Agent (Windows-Legacy) (Formerly StellarProtect (Legacy Mode))	Stellar Agent Linux (Formerly StellarProtect Linux)	StellarEnforce
	Discover Edition: 5.0	Discover Edition: 5.0		
3.3 Service Pack 2	3.3 Service Pack 2 and earlier versions	3.3 Service Pack 2 and earlier versions	1.0	N/A
3.3 Service Pack 1	3.3 Service Pack 1 and earlier versions	3.3 Service Pack 1 and earlier versions	1.0	N/A
3.3	3.3 and earlier versions	3.3 and earlier versions	1.0	N/A
3.2 Service Pack 2	3.2 Service Pack 2 and earlier versions	3.2 Service Pack 2 and earlier versions	N/A	N/A
3.2 Service Pack 1	3.2 Service Pack 1 and earlier versions	3.2 Service Pack 1 and earlier versions	N/A	N/A
3.2	3.2 and earlier versions	3.2 and earlier versions	N/A	N/A
3.1 Patch 1	3.1 Patch 1 and earlier versions	3.1 Patch 1 and earlier versions	N/A	N/A
3.1	3.1 and earlier versions	3.1 and earlier versions	N/A	N/A
3.0 Service Pack 1	3.0 Service Pack 1 and earlier versions	3.0 Service Pack 1 and earlier versions	N/A	N/A
3.0	3.0 and earlier versions	3.0 and earlier versions	N/A	N/A

Table 11. Supported Agent Version (continued)

Server Version	Agents Version			
StellarOne	Stellar Agent (Windows) (Formerly StellarProtect)	Stellar Agent (Windows-Legacy) (Formerly StellarProtect (Legacy Mode))	Stellar Agent Linux (Formerly StellarProtect Linux)	StellarEnforce
2.2	2.2 and earlier versions	1.5 and earlier versions	N/A	N/A
2.1	2.1 and earlier versions	1.4 and earlier versions	N/A	N/A
2.0	2.0 and earlier versions	1.3 and earlier versions	N/A	1.3 and earlier versions
1.2	1.2 and earlier versions	N/A	N/A	1.2 and earlier versions
1.1	1.1 and earlier versions	N/A	N/A	1.1 and earlier version
1.0	1.0	N/A	N/A	1.0

**Important:**

Please try to keep or upgrade the managed agents in or to the corresponding StellarOne major release version as indicated in the table above. Though StellarOne provides backward compatibility to support agents with earlier versions, new features or enhanced functionalities may not be applicable on some agents with earlier versions.

**Note:**

The StellarEnforce was renamed StellarProtect (Legacy Mode) upon the release of version 1.3.

Instance Data Encryption

You can encrypt the data associated with your StellarOne instance deployed on VMware ESXi, AWS EC2, or Microsoft Azure platform. Refer to [VMware ESXi Encryption Options](#), [Amazon EBS Encryption](#), or [Azure Disk Encryption](#) for more detailed procedures.



Note:

Since Microsoft has yet to support the encryption of Linux virtual machine run on Windows Hyper-V system, the StellarOne instance deployed on Hyper-V cannot be encrypted.

Before you begin the encryption configuration for StellarOne, verify the following requirements are met.

- Ensure that the virtual machine is powered off.
 - Ensure that you have the administrator or cryptographic operations privilege.
-



Important:

If the encrypted StellarOne instance requires technical support from TXOne Networks, ensure that you decrypt it for problem investigation.

Chapter 3. Installation

This chapter guides you through TXOne StellarOne installation. StellarOne is packaged in an Open Virtual Appliance (OVA) or Virtual Hard Disk v2 (VHDX) format and supports 5 types of platforms: VMware ESXi, VMware Workstation, Windows Hyper-V systems, AWS EC2 and KVM.

Topics in this chapter include:

- [StellarOne Installation Flow \(on page 32\)](#)
- [StellarOne Default Login Account \(on page 32\)](#)
- [StellarOne Onboarding to VMware ESXi \(on page 32\)](#)
- [StellarOne Onboarding to Windows Hyper-V \(on page 38\)](#)
- [StellarOne Onboarding to AWS EC2 \(on page 50\)](#)
- [StellarOne Onboarding to Azure \(on page 59\)](#)
- [StellarOne Onboarding to KVM \(on page 60\)](#)
- [Opening StellarOne Management Console \(on page 66\)](#)

StellarOne Installation Flow

Installing StellarOne web console requires performing the following steps:

1. Deploy a StellarOne instance on VMware ESXi or Workstation, Windows Hyper-V, AWS EC2 platform, or Microsoft Azure.
2. Add an external hard disk with at least 70 GB of space to the StellarOne instance.
3. For VMware or Hyper-V system, launch the StellarOne instance, and then run the OOB process and configure settings such as IP address and communication ports in the setup CLI.
4. Log on StellarOne web console to set up the administrator's account.
5. Log on StellarOne web console to activate the product license and set time properties.

StellarOne Default Login Account

For new customers using StellarOne 3.3 or later, the following default login account applies.

Windows Hyper-V, VMware ESXi, Azure and KVM

Default login account: txuser

AWS EC2

Default login account: admin



Note: For existing customers upgrading to StellarOne 3.3 or later from a version prior to 3.3, the above default login account **does not apply**.

StellarOne Onboarding to VMware ESXi

This section describes how to deploy StellarOne to a VMware ESXi system.

Prerequisites

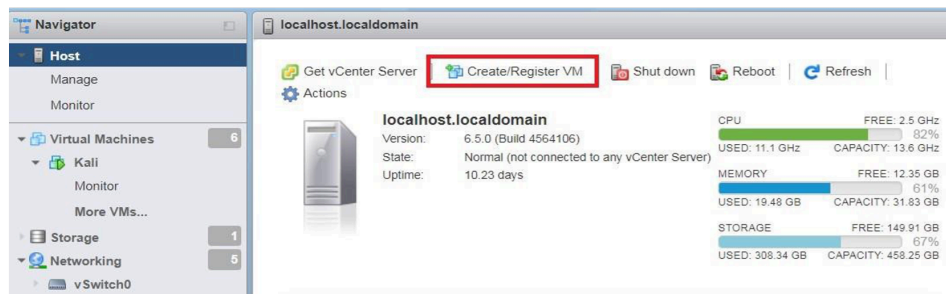
- The OVA packages provided by TXOne must be available and accessible to VMware ESXi.
- VMware ESXi 6.5 or above is required.
- The necessary networks have been properly created for ESXi.
- An external disk with at least 70 GB.

Deploying StellarOne on the VMware ESXi

The following section describes the procedures of deploying StellarOne from an OVA file to the VMware ESXi system.

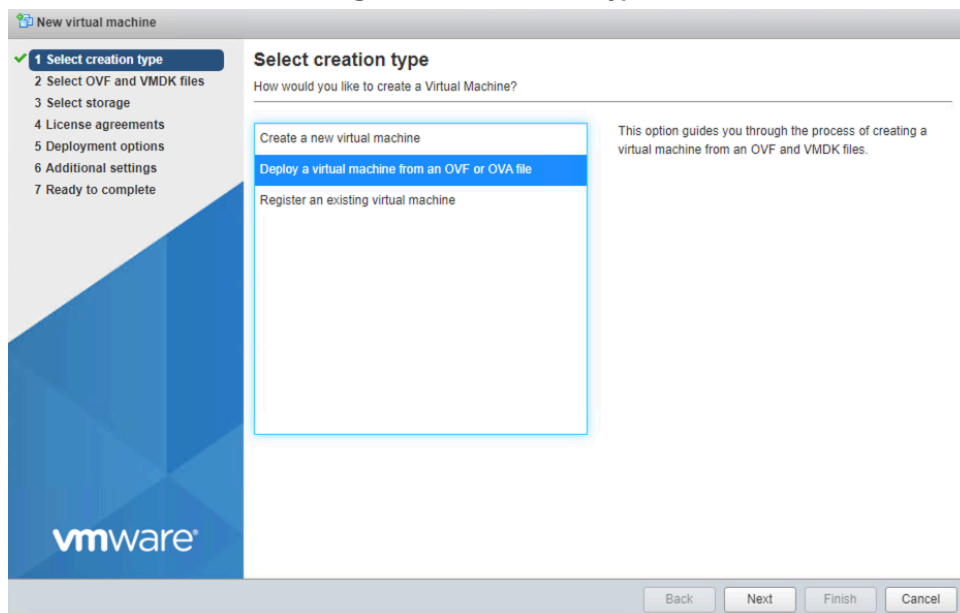
1. Log into the VMware vSphere web client.
2. Under **Navigator**, click **Host > Create/Register VM**.

Figure 1. Navigator



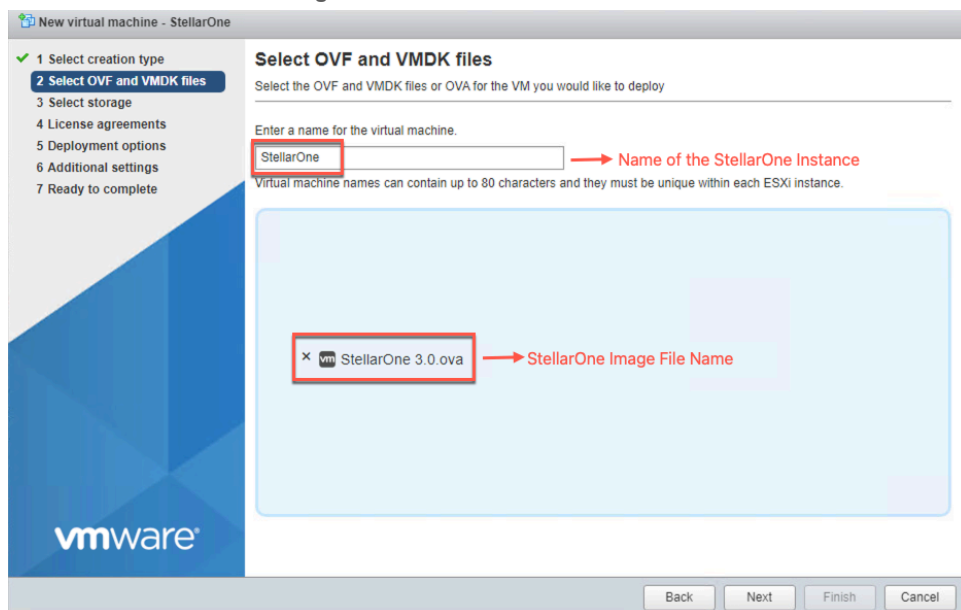
3. In **Select creation type**, select **Deploy a virtual machine from an OVF or OVA file** and click **Next**.

Figure 2. Select creation type



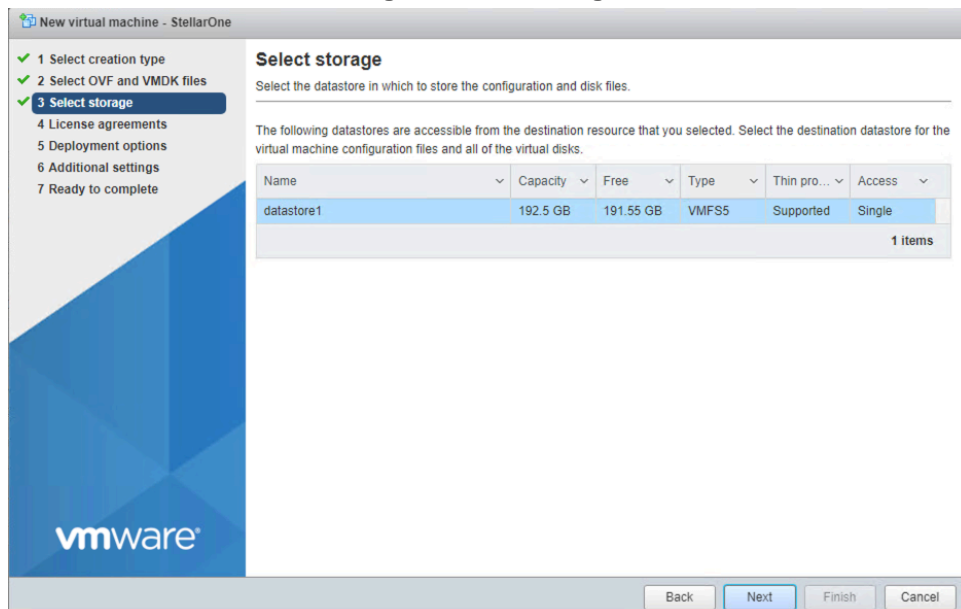
4. Specify a name for your new StellarOne instance and select the StellarOne disk image to upload.

Figure 3. Select OVF and VMDK files



5. Choose a storage location for the StellarOne instance and click **Next**.

Figure 4. Select storage



6. Select deployment options and click **Next**.

Figure 5. Deployment options

The screenshot shows the 'New virtual machine - StellarOne' wizard. On the left, a progress bar indicates the steps: 1 Select creation type, 2 Select OVF and VMDK files, 3 Select storage, 4 Deployment options (highlighted), and 5 Ready to complete. The main area is titled 'Deployment options' with the instruction 'Select deployment options'. It contains two sections: 'Network mappings' with a dropdown menu set to 'NAT' and 'VM Network', and 'Disk provisioning' with radio buttons for 'Thin' (selected) and 'Thick'. At the bottom, there are 'Back', 'Next', 'Finish', and 'Cancel' buttons.

7. When you see **Ready to complete**, click **Finish** to start the deployment.

Figure 6. Ready to complete

The screenshot shows the 'New virtual machine - StellarOne' wizard at the 'Ready to complete' step. The progress bar on the left now highlights '5 Ready to complete'. The main area is titled 'Ready to complete' with the instruction 'Review your settings selection before finishing the wizard'. It contains a table summarizing the configuration:

Product	Unknown
VM Name	StellarOne
Disks	instance.vmdk
Datastore	datastore1
Provisioning type	Thin
Network mappings	NAT: VM Network
Guest OS Name	Debian_64

Below the table, there is a yellow warning icon and the text: 'Do not refresh your browser while this VM is being deployed.' At the bottom, there are 'Back', 'Next', 'Finish', and 'Cancel' buttons.

8. Under the **Recent Tasks** pane, you will see a progress bar indicating the StellarOne image is being uploaded. Please wait until the upload is finished.
9. Add an external disk with at least 70 GB of capacity to the StellarOne instance.

- a. Close the StellarOne instance if it is open.

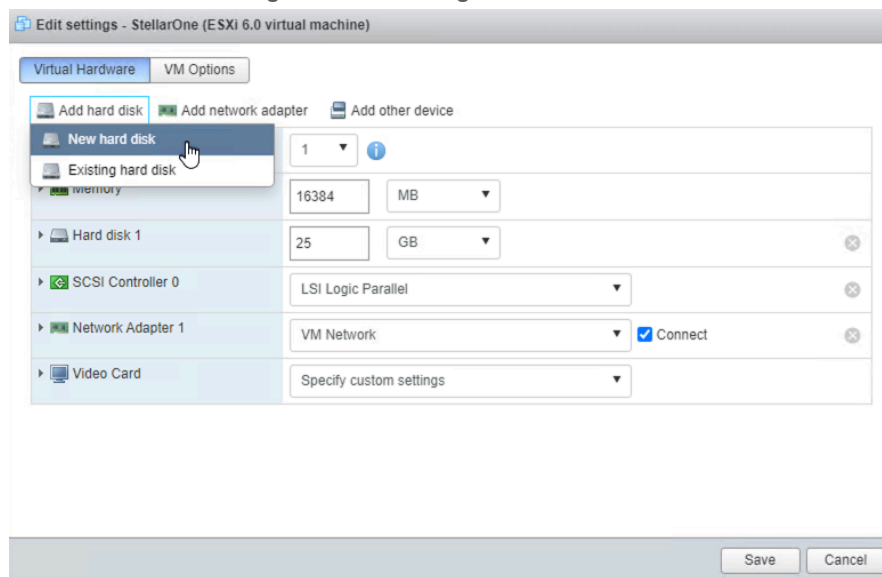


Note:

See [Hardware Requirements for Deploying StellarOne on VMware/Hyper-V/AWS EC2 \(on page 19\)](#) for the recommended 2nd disk size for StellarOne.

- b. Follow the procedures to add the external disk: **Actions > Edit settings > Add hard disk > New hard disk**

Figure 7. Edit settings - New hard disk



- c. Set the new hard disk space to 70 GB.
 - a. If you need to increase the number of logs StellarOne can store, follow the procedures.
 - i. Shut down StellarOne
 - ii. Increase the external disk capacity to fit the maximum log requirements
 - iii. Restart the StellarOne instance. After that, the storage space available for StellarOne log files will be expanded.
 - b. If you want to migrate the existing StellarOne settings to the newly launched virtual machine, see [System Migration \(on page 88\)](#).



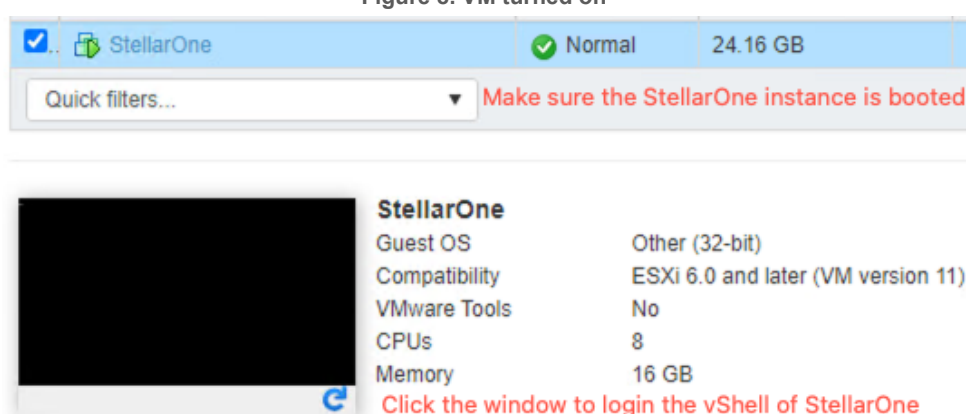
Note:



- a. StellarOne requires one external disk with minimum capacity of 70GB; otherwise, StellarOne will not finish initialization and will not complete the boot process.
- b. The external disk is used to store the system configurations and event logs. You may attach the external disk of a terminated StellarOne instance here instead of adding a new disk if you want to migrate the previous configurations and logs to a new instance.

10. Turn on the virtual machine.

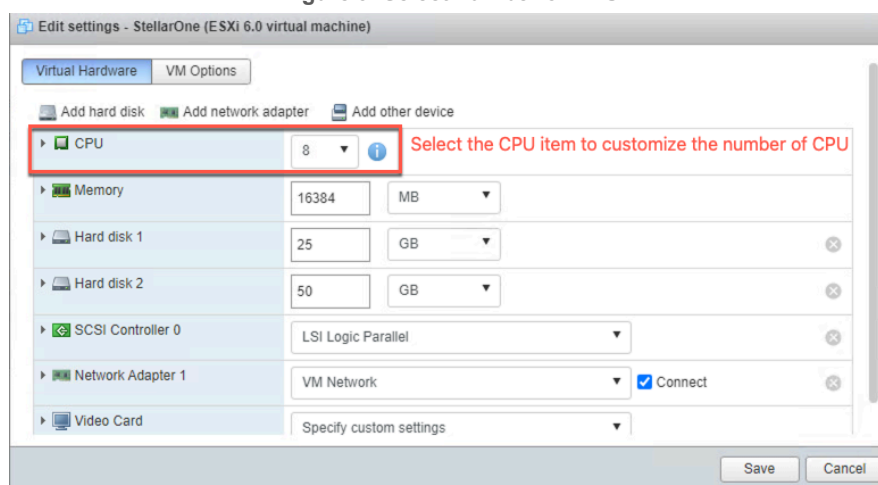
Figure 8. VM turned on



11. (Optional) Adjust your StellarOne instance to use the proper resource configurations based on the default setting of 8 CPU cores and 16 GB Memory.

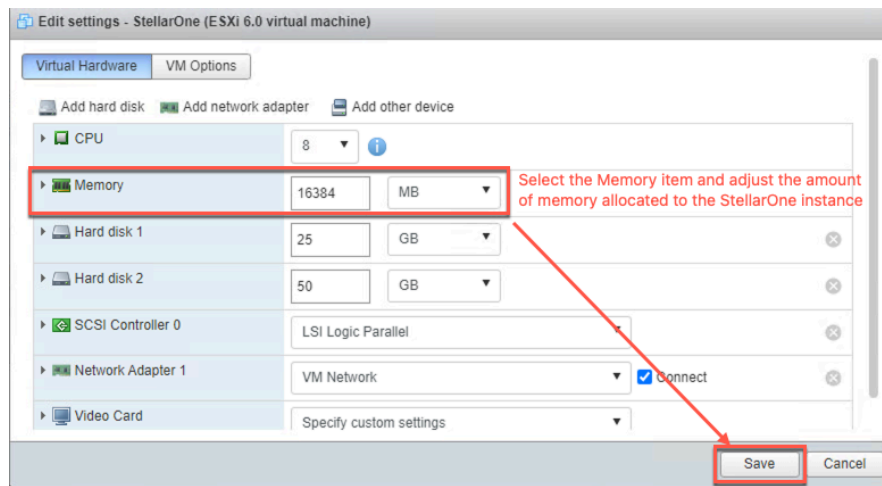
- a. Shut down the StellarOne instance and click **Actions > Edit settings**. The **Edit settings** window appears.
- b. Configure the number of CPU cores.

Figure 9. Select number of CPU



- c. Configure the amount of Memory.

Figure 10. Configure Memory

**Note:**

See the [Hardware Requirements for Deploying StellarOne on VMware/Hyper-V/AWS EC2 \(on page 19\)](#) for the CPU and memory requirements for agent deployment and corresponding StellarOne configuration and resource allocation.

- d. Boot the StellarOne instance.

StellarOne Onboarding to Windows Hyper-V

This section describes how to deploy StellarOne to the Windows Hyper-V system.

Prerequisites

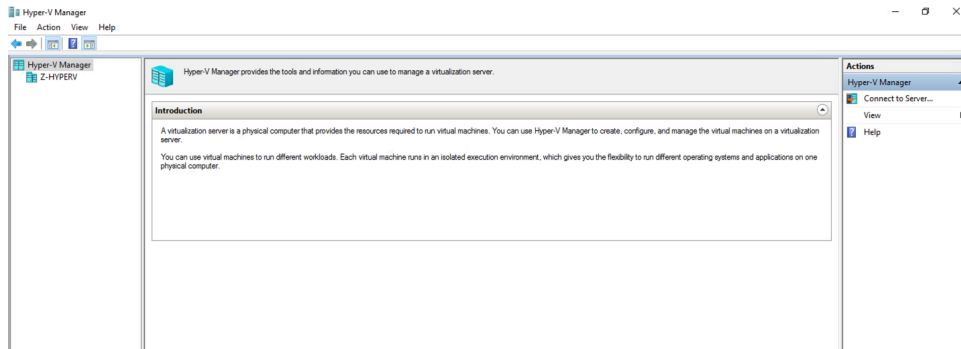
- The VHDX packages provided by TXOne must be available and accessible to Windows Hyper-V.
- Windows Server 2019, Hyper-V Manager Windows 10 or above.
- The necessary networks have been properly created for Windows Hyper-V.
- An external disk with at least 70 GB.

Deploying StellarOne to a Hyper-V System

The following section describes the procedures of deploying StellarOne from a VHDX File to a Hyper-V system.

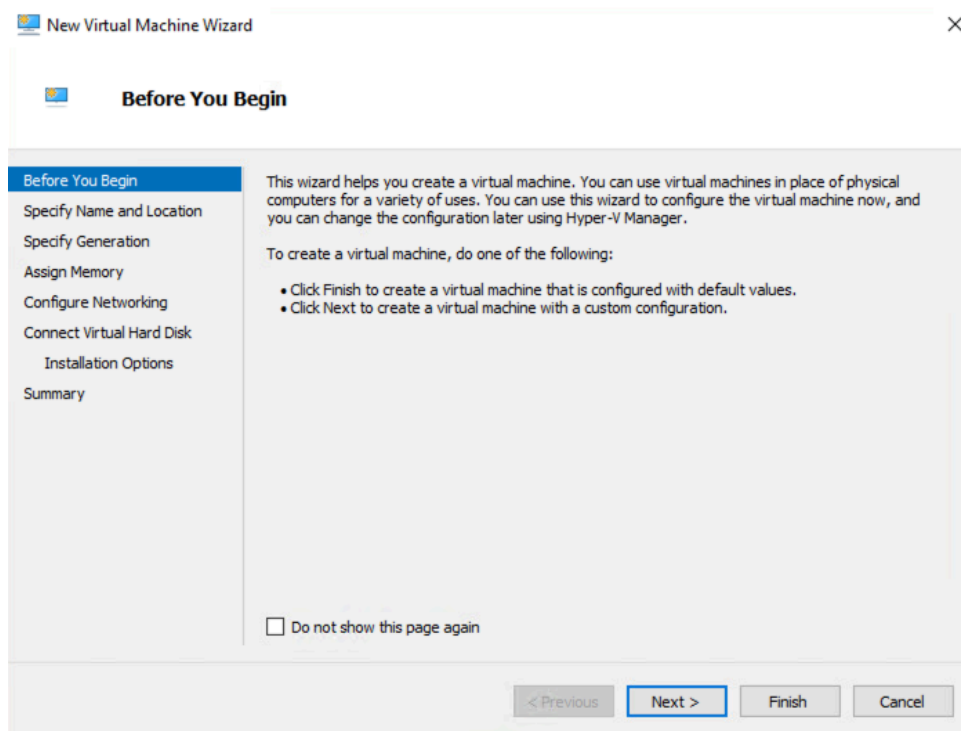
1. Launch **Hyper-V Manager**

Figure 11. Hyper-V Manager



2. Under **Actions**, click **New > Virtual Machine**.
3. The **New Virtual Machine Wizard** appears, click **Next**.

Figure 12. New Virtual Machine Wizard: Before You Begin



4. In **Specify Name and Location**, type a name for your new virtual machine and click **Next**.

Figure 13. New Virtual Machine Wizard: Specify Name and Location

Specify Name and Location

Choose a name and location for this virtual machine.

The name is displayed in Hyper-V Manager. We recommend that you use a name that helps you easily identify this virtual machine, such as the name of the guest operating system or workload.

Name:

You can create a folder or use an existing folder to store the virtual machine. If you don't select a folder, the virtual machine is stored in the default folder configured for this server.

☐ Store the virtual machine in a different location

Location:

! If you plan to take checkpoints of this virtual machine, select a location that has enough free space. Checkpoints include virtual machine data and may require a large amount of space.

< Previous **Next >** Finish Cancel

5. In **Specify Generation**, select **Generation 1**.

Figure 14. New Virtual Machine Wizard: Specify Generation

Specify Generation

Choose the generation of this virtual machine.

☒ **Generation 1**
This virtual machine generation supports 32-bit and 64-bit guest operating systems and provides virtual hardware which has been available in all previous versions of Hyper-V.

☐ **Generation 2**
This virtual machine generation provides support for newer virtualization features, has UEFI-based firmware, and requires a supported 64-bit guest operating system.

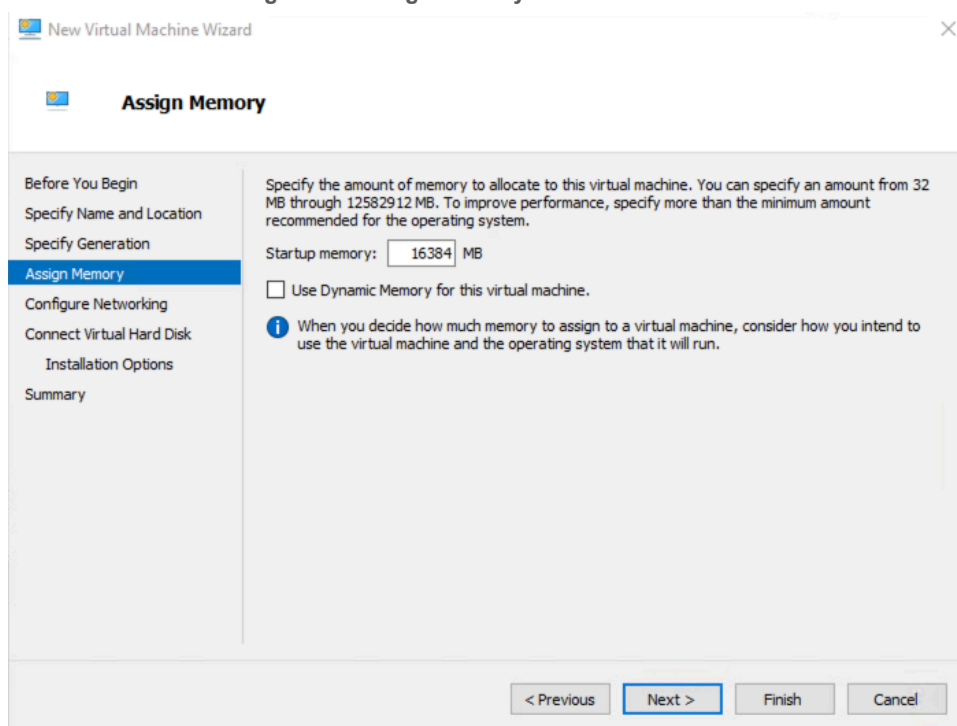
! Once a virtual machine has been created, you cannot change its generation.

[More about virtual machine generation support](#)

< Previous **Next >** Finish Cancel

6. In **Assign Memory**, allocate memory for the new virtual machine and click **Next**.

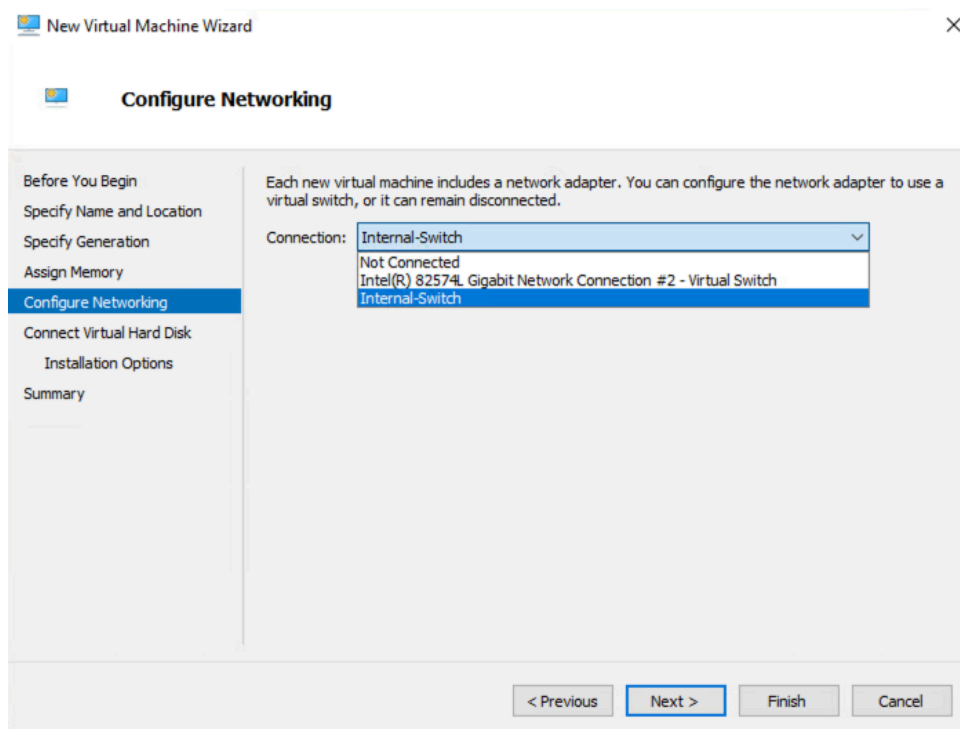
Figure 15. Assign Memory for Virtual Machine

**Note:**

For further agent deployment and configurations, it is recommended to at least meet the hardware requirements: 8 CPU cores and 16 GB Memory.

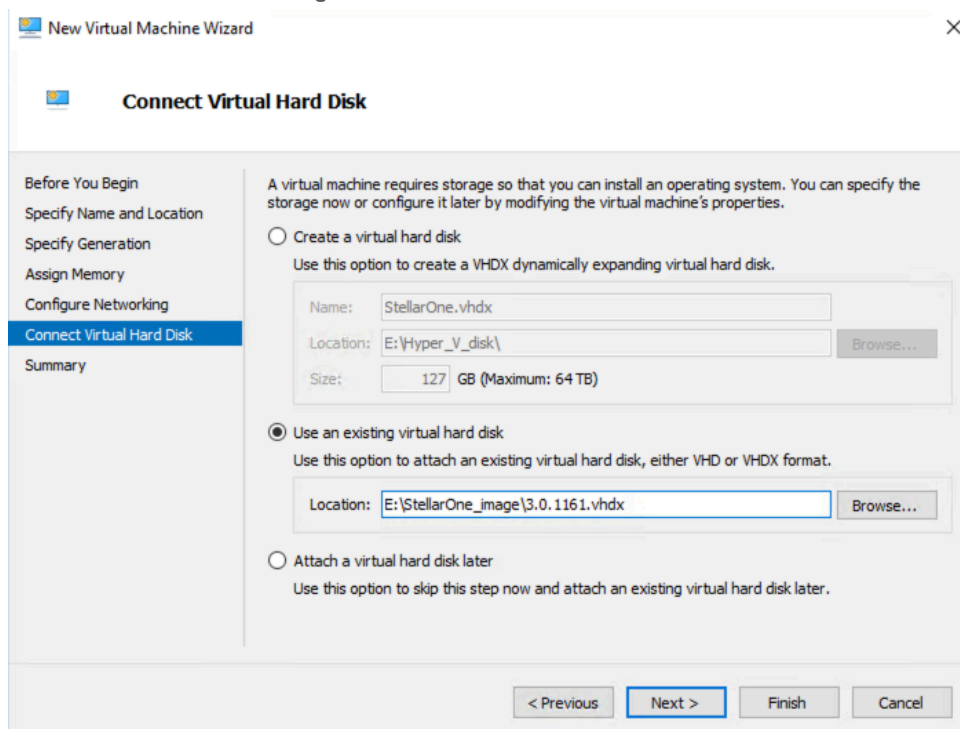
7. Configure the network settings for the new virtual machine, and then click **Next**.

Figure 16. Configure Networking for Virtual Machine



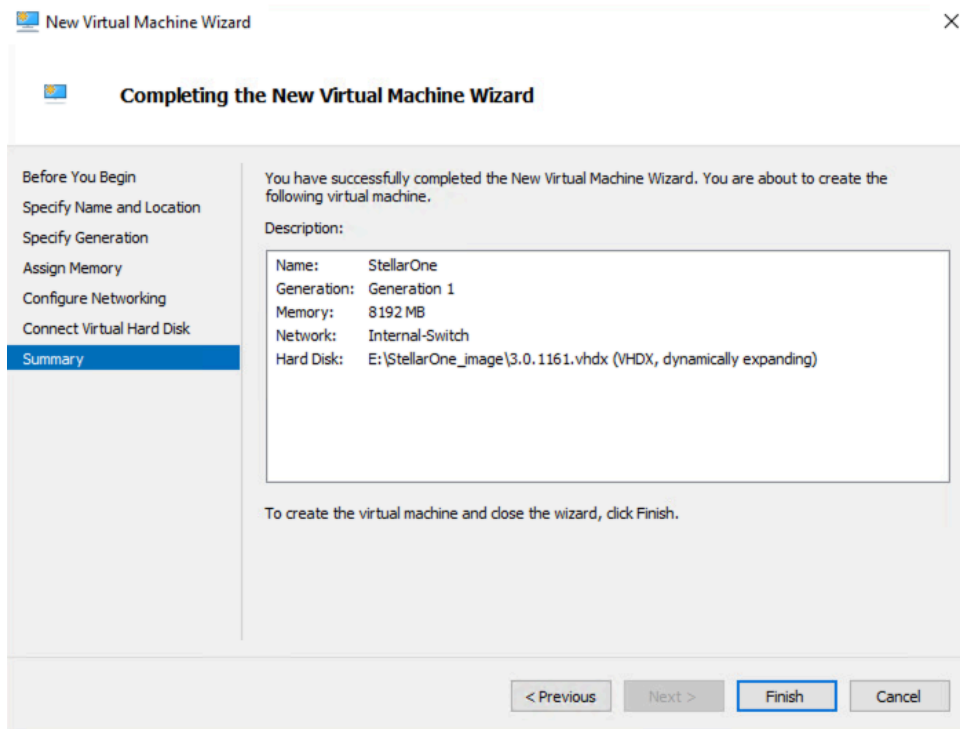
8. Select a virtual hard disk (the StellarOne .vhdx file) and click **Next**.

Figure 17. Connect Virtual Hard Disk



9. Check your settings and click **Finish**.

Figure 18. Completing the New VM Wizard



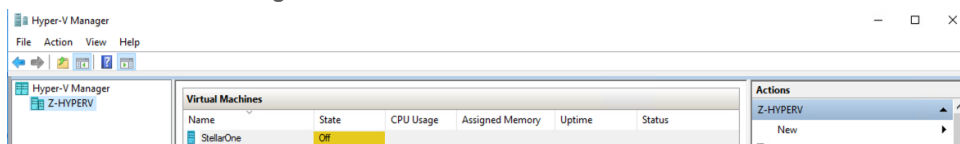
10. Add a new disk for the StellarOne virtual machine.



Note:

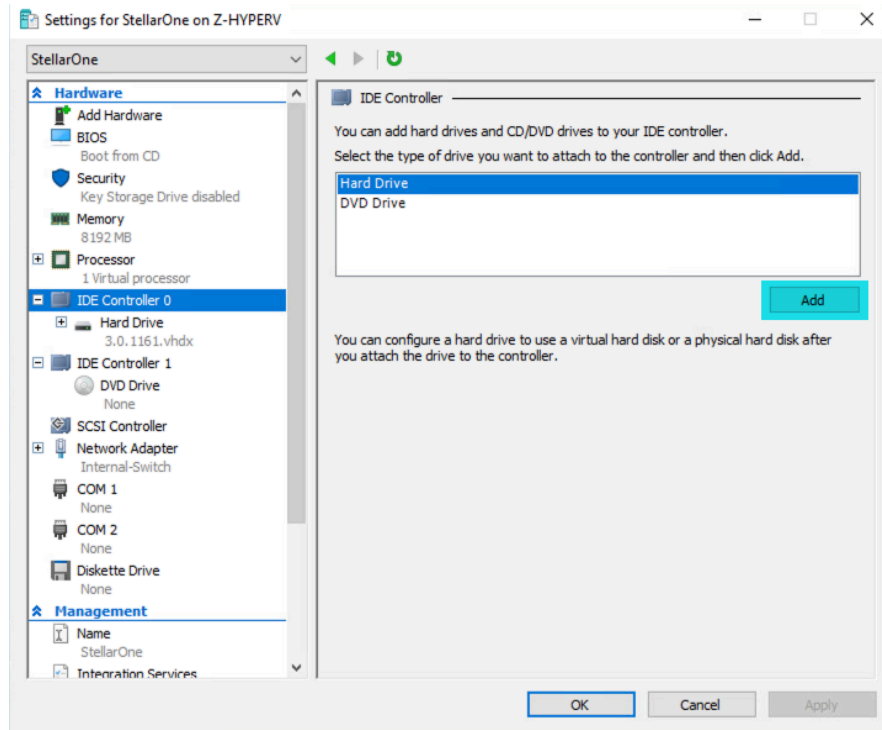
Make sure the previous StellarOne instance is turned off.

Figure 19. State of StellarOne instance is off



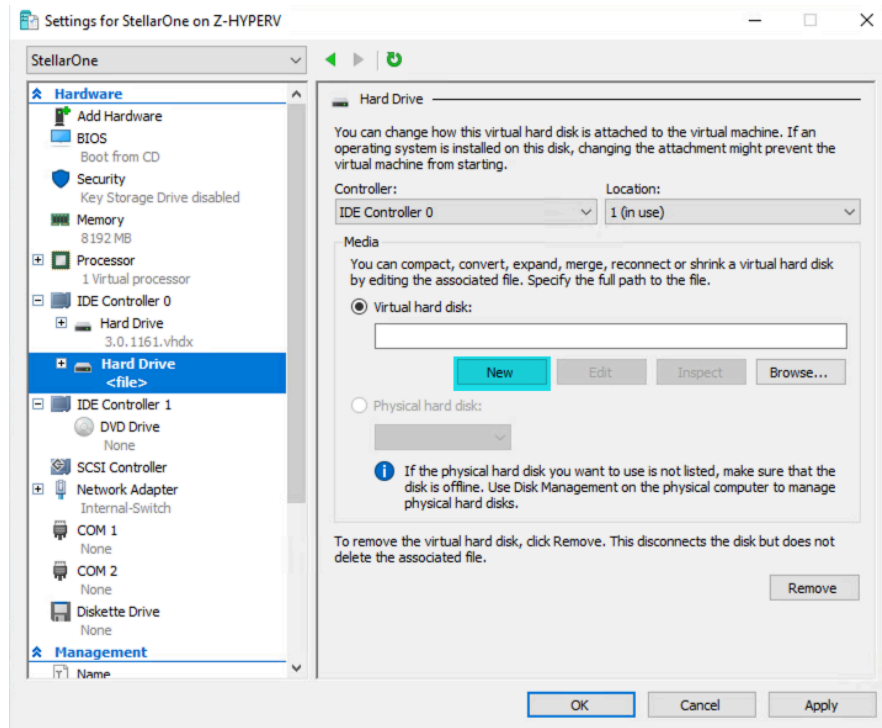
- a. Select the StellarOne virtual machine and right click to select **Settings** from the context menu.
- b. Select **Hard Drive** from the **IDE Controller 0** item and click **Add**.

Figure 20. Settings for StellarOne - 1



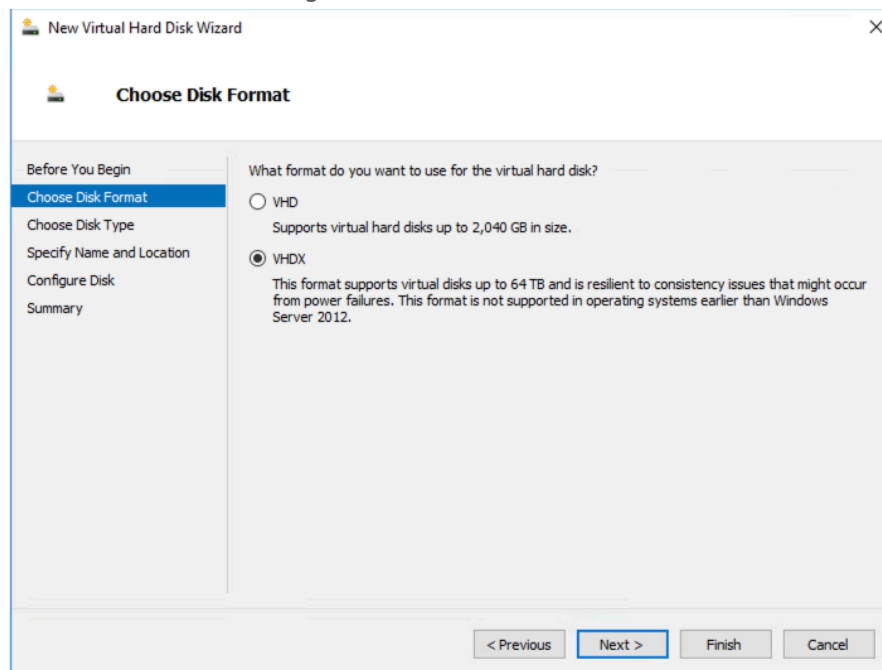
- c. Click **New**.

Figure 21. Settings for StellarOne - 2



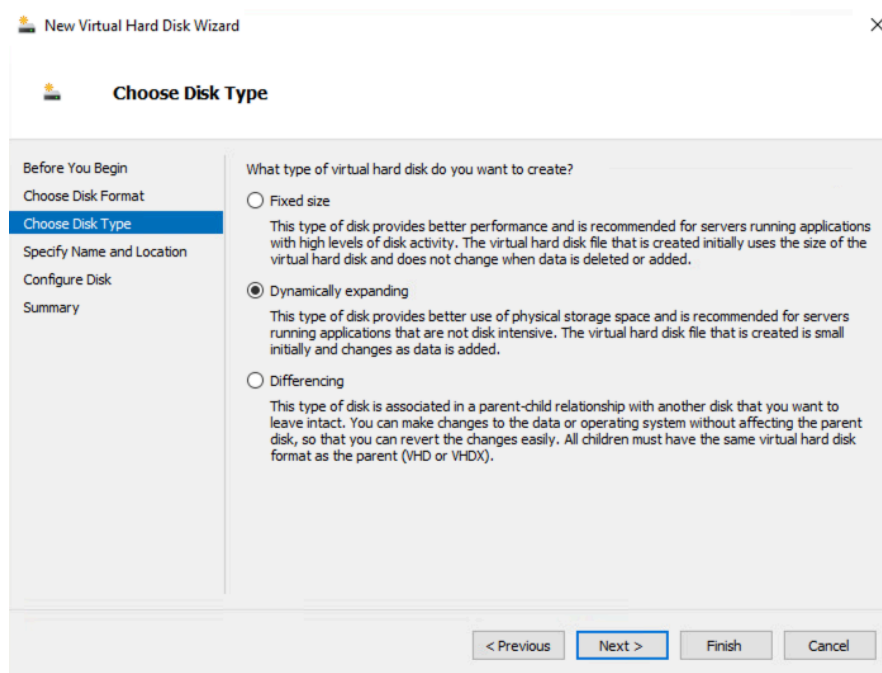
- d. The **New Virtual Hard Disk Wizard** appears. Click **Next**.
- e. In **Choose Disk Format**, select **VHDX** as the disk format and click **Next**.

Figure 22. Choose Disk Format



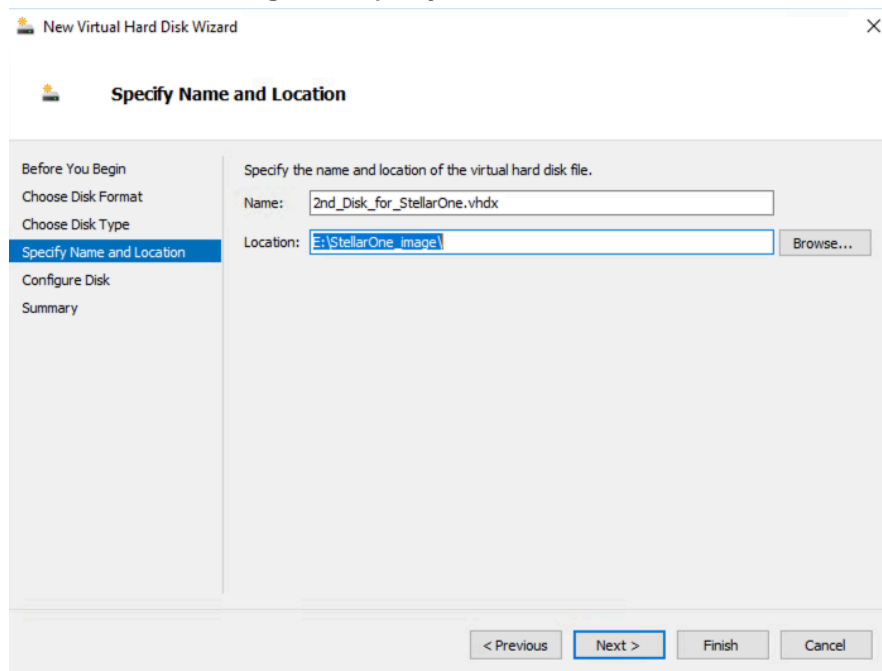
- f. In **Choose Disk Type**, select **Dynamically expanding** as the disk type and click **Next**.

Figure 23. Choose Disk Type



- g. Specify the name and location of the virtual hard disk file.

Figure 24. Specify Name and Location



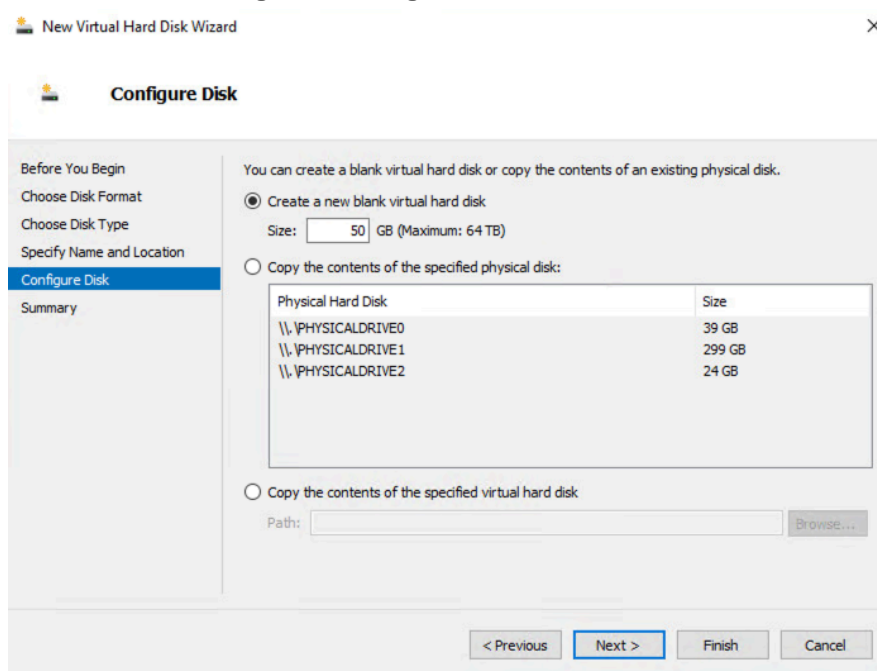
- h. Configure disk size.



Note:

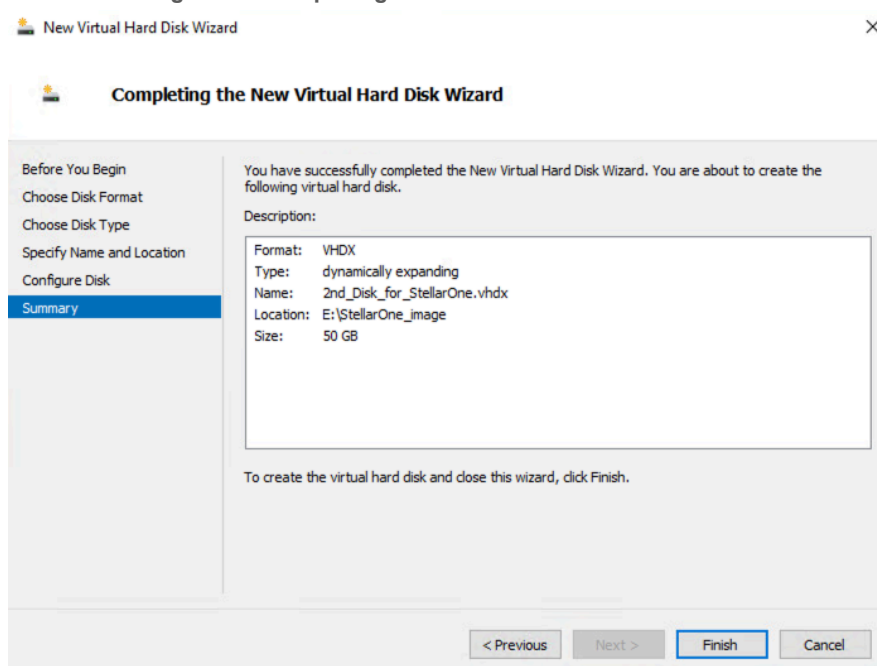
 See [Hardware Requirements for Deploying StellarOne on VMware/Hyper-V/AWS EC2 \(on page 19\)](#) for the recommended 2nd disk size for StellarOne.

Figure 25. Configure Disk for StellarOne



- i. Click **Next** to check your settings.

Figure 26. Completing the New Virtual Hard Disk Wizard

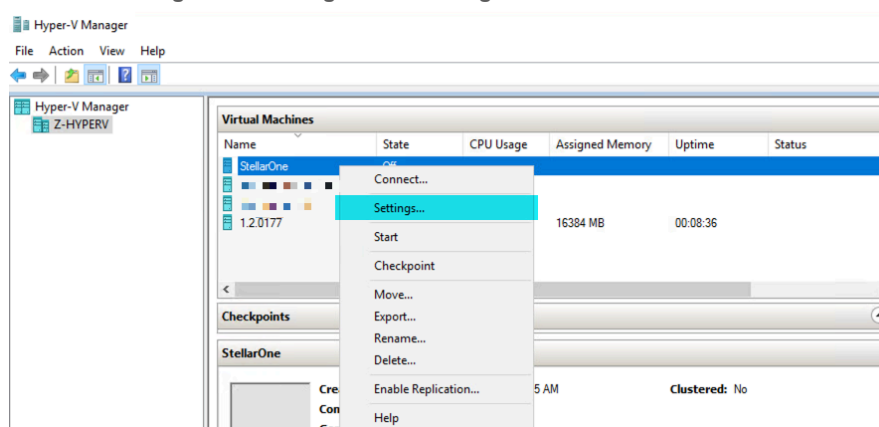


- j. Click **Finish** to complete the settings.

11. (Optional) See [Hardware Requirements for Deploying StellarOne on VMware/Hyper-V/AWS EC2 \(on page 19\)](#) for the CPU and memory requirements for agent deployment and corresponding StellarOne configuration and resource allocation. It is recommended to at least meet the hardware requirements (4 CPU cores, 16 GB Memory).

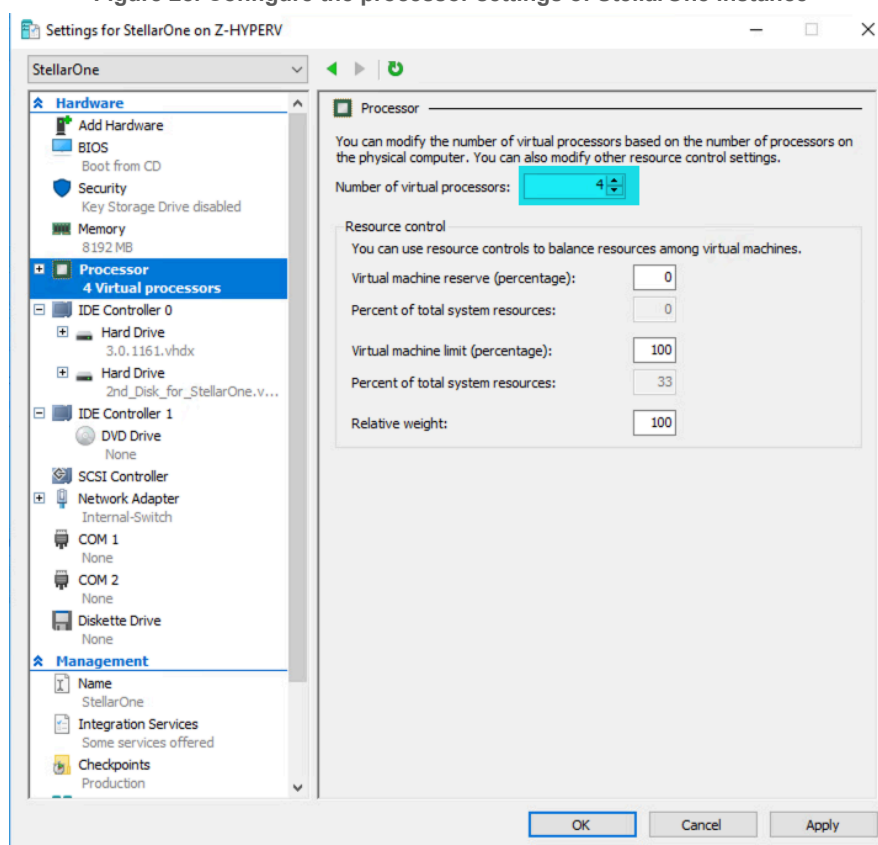
- a. Shut down the StellarOne instance. Select and right click the instance, and then click **Settings**.

Figure 27. Configure the settings of StellarOne instance



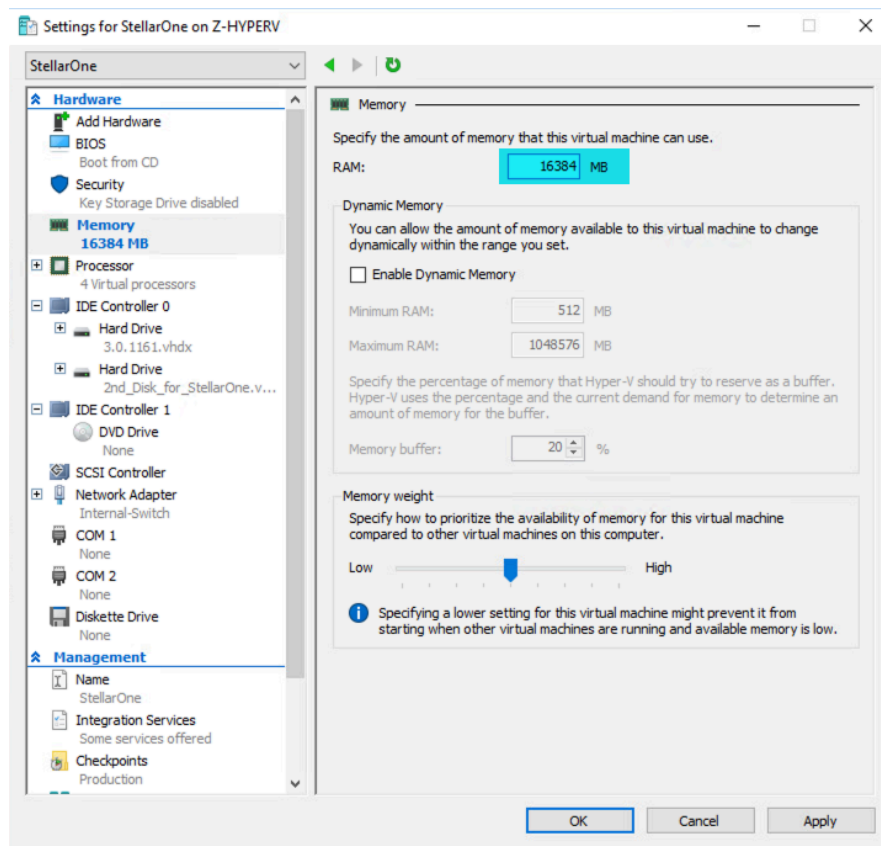
- b. In **Processor**, configure the number of virtual processors and the associated resource control settings. Click **OK** to complete the settings.

Figure 28. Configure the processor settings of StellarOne instance



- c. In the **Memory** section, specify the amount of memory that the StellarOne instance can use.

Figure 29. Configure the memory settings of StellarOne instance



d. Boot the StellarOne instance.

StellarOne Onboarding to AWS EC2

This section describes how to deploy StellarOne on the AWS EC2 platform.

Prerequisites

- An AWS account is required
- StellarOne for AWS supports only BYOL (Bring Your Own License)
- An external disk (EBS) with at least 70 GB.



Note:

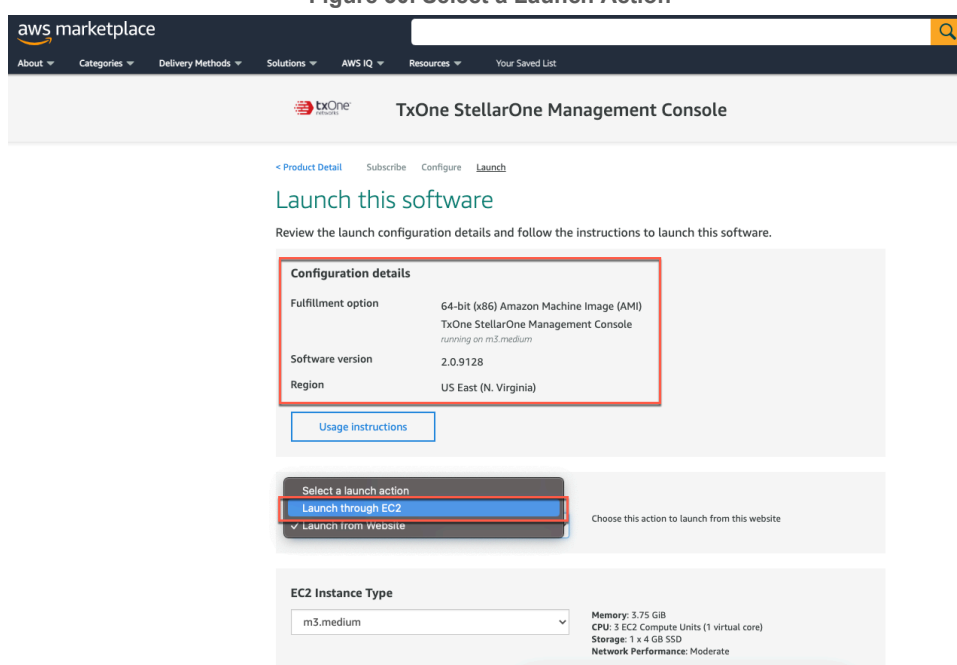
Please contact your support provider for the BYOL license.

Deploying StellarOne on AWS EC2

Below section details procedures of deploying StellarOne from BYOL AMI on the AWS EC2 platform.

1. Go to the **AWS Marketplace** homepage.
2. Enter the search string such as TXOne or StellarOne in the search bar and then run the search.
3. Click the search result and read the product information carefully before proceeding to the subscription process.
4. After accepting the terms and conditions for using StellarOne, choose the **Fulfillment option**, **Software version**, and **Region** to launch StellarOne.
5. Select **Launch through EC2** as the launch action.

Figure 30. Select a Launch Action



6. Log on the AWS EC2 console.
7. Go to **Images > AMIs**.
8. Select the region you chose in step 4.
9. Find the target AMI from the list of **AMI ID**.
10. Select the target AMI and click **Launch Instance from AMI**.
11. Select a supported instance type.

Figure 31. Select an Instance Type

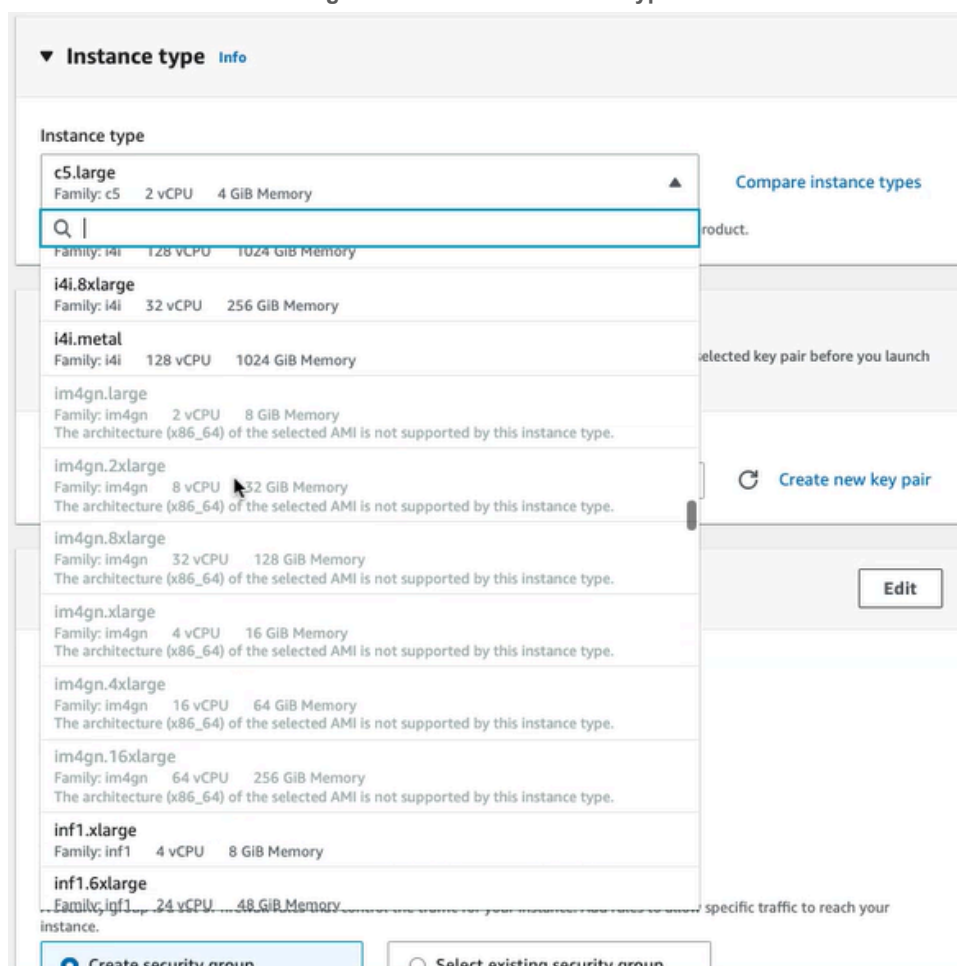
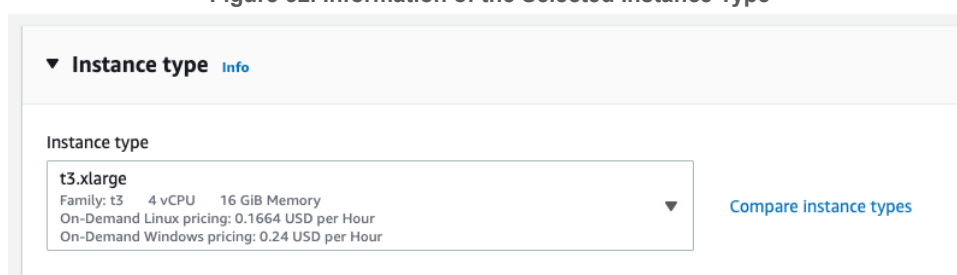


Figure 32. Information of the Selected Instance Type

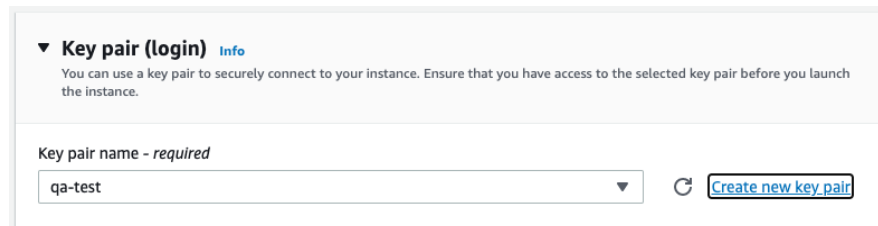
**Note:**

- The instance types that do not meet StellarOne's deployment requirements will be unavailable for selection.
- See [Hardware Requirements for Deploying StellarOne on VMware/Hyper-V/AWS EC2 \(on page 19\)](#) for determining which instance type to use.

12. Configure the instance settings:

a. Select or create the **Key pair (login)**

Figure 33. Key pair (login)



▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

qa-test ▼

↻ [Create new key pair](#)

b. Configure the **Network settings**:

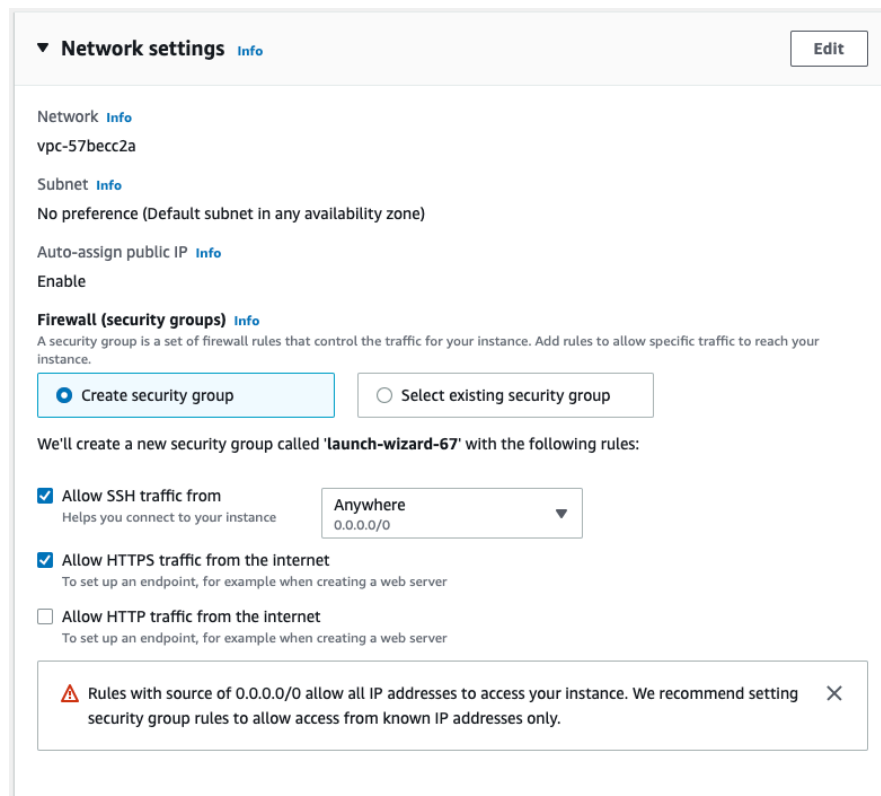
- Be sure to create security group for allowing specific data exchanges to access your instance. It is required to enable the **HTTPS traffic from the Internet** to allow StellarOne to manage endpoints on the network.
- (Optional) If you have the need for SSH login, you can also enable the **Allow SSH traffic from** and select **Anywhere** or specify the IP address.

**Note:**

See [Outbound Ports and FQDN Used \(on page 23\)](#) for configuring the ports that should have access to StellarOne.

- Be sure to grant 8000 or 9443, the dedicated port for Stellar Agent (Windows) or Stellar Agent (Windows-Legacy), access to your instance.
- For security reasons, it is recommended to allow 443 or 22, the web port or SSH port for StellarOne, to be accessible from trusted IP address.

Figure 34. Network settings



▼ **Network settings** [Info](#) Edit

Network [Info](#)
vpc-57becc2a

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-67' with the following rules:

☒ Allow SSH traffic from Anywhere
0.0.0.0/0
Helps you connect to your instance

☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

 Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only. ×

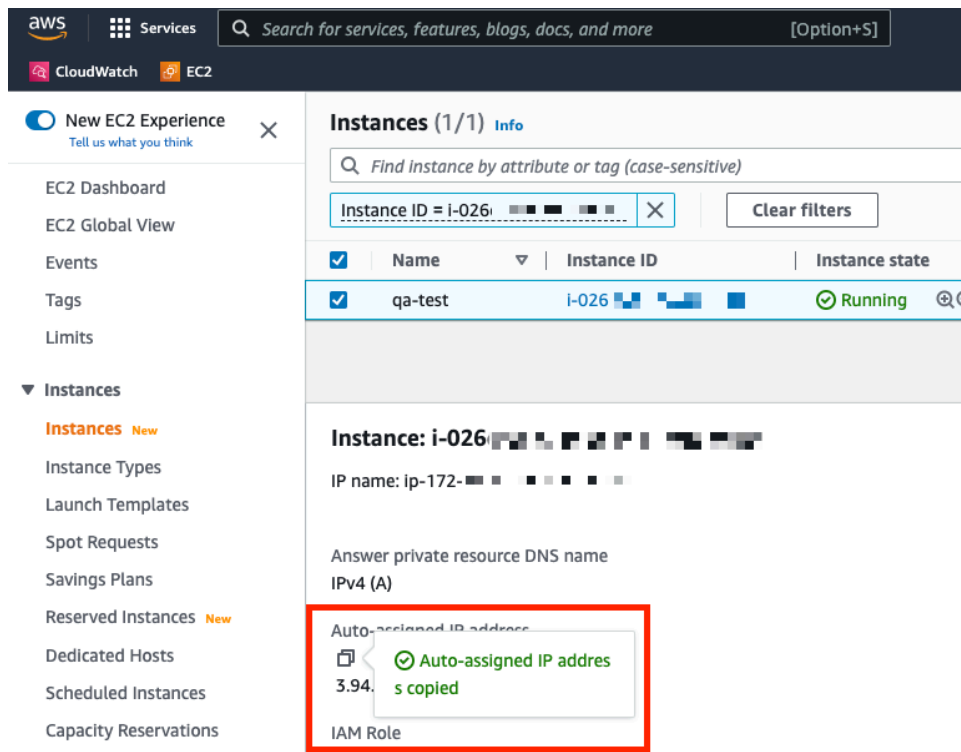
- c. Add an EBS with at least 70 GB of space to the StellarOne instance in **Configure storage**.
- d. Click **Launch Instance**.

**Note:**

It may take 5 to 10 minutes to complete the deployment.

13. Find the StellarOne instance and copy its assigned IP address.

Figure 35. Auto-assigned IP address

**Note:**

The auto-assigned IP address may change if the instance has been rebooted. See [Associating the Elastic IP Address with an Instance \(on page 57\)](#) for assigning a static IPv4 address to your instance.

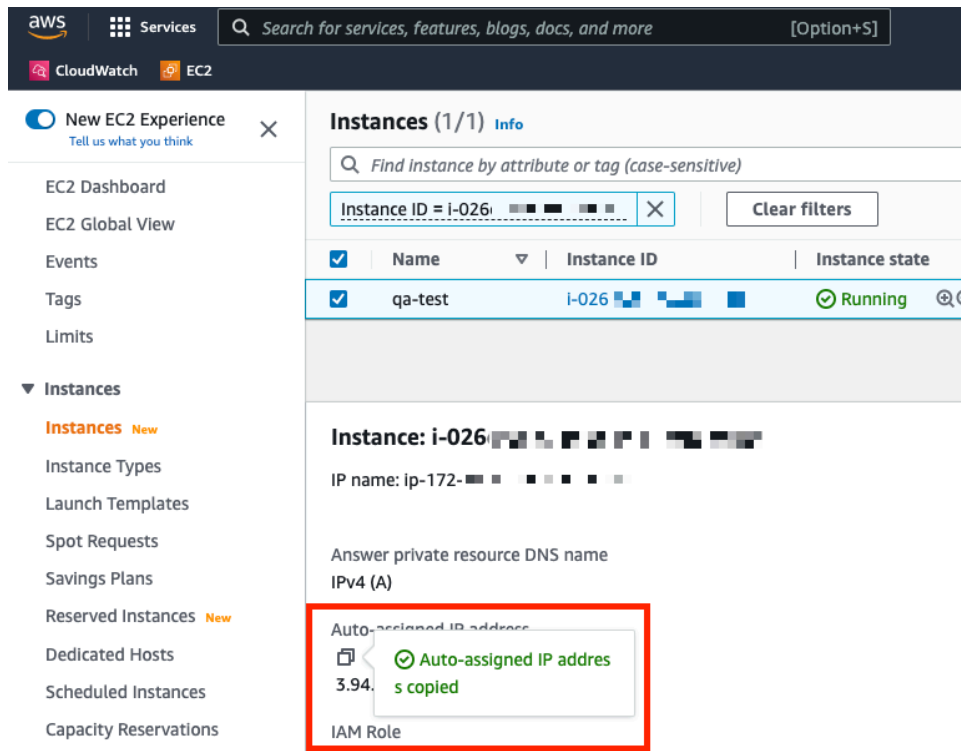
14. See [Opening StellarOne Management Console \(on page 66\)](#) for logging on StellarOne via a web browser, or [Accessing StellarOne via SSH \(on page 55\)](#) for accessing StellarOne via SSH.

Accessing StellarOne via SSH

This section describes how to access StellarOne via SSH.

1. Find the StellarOne instance on the AWS EC2 and copy its auto-assigned IP address.

Figure 36. Auto-assigned IP address



2. Open the SSH terminal on your device and run the following command: `ssh -i <private key>.pem admin@<auto-assigned IP address>`



Note:

The auto-assigned IP address may change if the instance has been rebooted. Please refer to [Associating the Elastic IP Address with an Instance \(on page 57\)](#) for assigning a static IPv4 address to your instance.



Important:



- Ensure that you use `admin` as the SSH login username.
- If you access StellarOne via SSH, it is not required to go through the OOB process and change password.

3. Check if the service of StellarOne instance is activated and running.

Figure 37. Check Service Status via SSH

```
$ env ls
Hostname: ODC
Status: RUNNING
Product Serial Number: 7a [redacted] a9
Version: 2.2.1148
Advertise Address: Not Set
DPI Engine Version:
DPI Pattern Version:
StellarProtect (Legacy Mode) Agent Up Port:8000
StellarProtect (Legacy Mode) Agent Down Port:14336
StellarProtect Agent Up Port: 9443
StellarProtect Agent Down Port:14336
Locale: en
```

Associating the Elastic IP Address with an Instance

The following section describes how to assign the Elastic IP address to your StellarOne instance deployed from the AMI on AWS EC2 platform.



Note:

The Elastic IP address is a static IPv4 address designed for dynamic cloud computing, and is allocated to your AWS account.

1. Log on the AWS EC2 console and go to **Network & Security > Elastic IPs**.
2. Select the Elastic IP address to associate and choose **Actions > Associate Elastic IP address**.
3. For **Resource type**, select **Instance**.
4. Choose the target instance.



Note:

You can search for a specific instance by typing relevant strings in the search bar.

5. (Optional) For **Private IP address**, specify a private IP address with which to associate the Elastic IP address.
6. Click **Associate**.

Figure 38. Associate the Elastic IP Address with an Instance

aws Services Search for services, features, blogs, docs, and more [Option+S]

CloudWatch EC2

EC2 > Elastic IP addresses > Associate Elastic IP address

Associate Elastic IP address

Choose the instance or network interface to associate to this Elastic IP address (35.168.1.10)

Elastic IP address: 35.168.1.10

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance

☐ Network interface

Warning If you associate an Elastic IP address to an instance that already has an Elastic IP address associated, this previously associated Elastic IP address will be disassociated but still allocated to your account. [Learn more](#)

Instance

i-026c...

Private IP address
The private IP address with which to associate the Elastic IP address.

Choose a private IP address

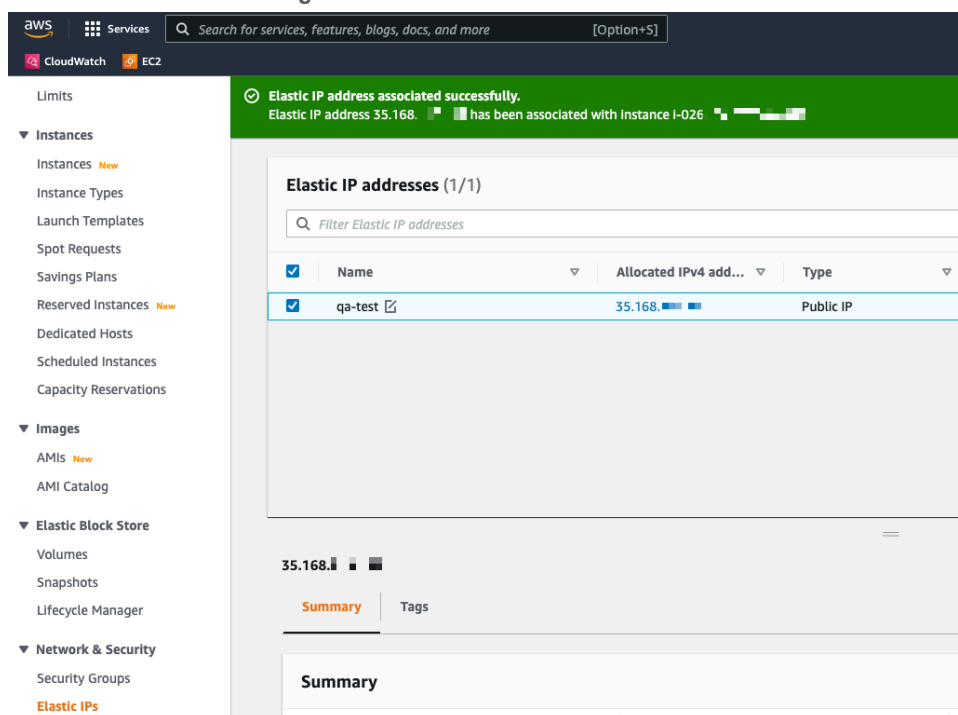
Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.

☐ Allow this Elastic IP address to be reassociated

Cancel Associate

7. A message appears indicating the Elastic IP address has been associated to the target instance.

Figure 39. Associate Elastic IP Address



8. You can use the Elastic IP address to log on StellarOne via a web browser or via SSH now.
9. (Optional) Choose one of the methods below to access StellarOne via SSH with the Elastic IP address:

- Replace the advertise address with the Elastic IP address by typing:

```
env advertise <the elastic IP address>
```
- Replace the advertise address with the Elastic Load Balancer address by typing:

```
env advertise <the ELB address>
```

StellarOne Onboarding to Azure

This section describes how to deploy StellarOne on the Microsoft Azure platform.

Prerequisites

- A Microsoft Azure account

Deploying StellarOne on Azure

The below instructions detail the general steps required to deploy StellarOne on Microsoft Azure. Other than StellarOne specific information, such as minimum requirements and post-deployment configuration, the setup is standard. As Azure's user interface changes frequently, instead of giving step-by-step instructions that may become quickly outdated, we have instead given links to the official Azure documentation for steps that may require further guidance.

1. Prepare and Import the VHD

- a. Extract the provided .vhd.zip archive to obtain the .vhd image file.
- b. Upload the .vhd image file to your [Azure storage account](#)'s Blob container.
 - i. When uploading, set the blob type to **Page blob** (recommended for .vhd files).
 - ii. Follow the [official Azure documentation](#) for detailed instructions.

2. Create the Image

- a. In the Azure portal, create a custom image from the uploaded .vhd file in your Blob container. Use the following settings:
 - i. OS type: **Linux**
 - ii. VM generation: **Gen 1**
 - iii. Region: the same region as the storage account containing the uploaded .vhd file
 - iv. Storage blob: the .vhd file uploaded in the previous step
- b. Follow the [official Azure documentation](#) for detailed instructions.

3. Create the Virtual Machine

- a. Create a virtual machine from the image you created in the previous step, using the following settings:
 - i. Minimum CPU: 4 vCPU
 - ii. Minimum memory: 16 GB
 - iii. Authentication: only SSH public key authentication is supported.
 1. The default user account is `txuser`.
 2. Download and securely store the private key when prompted. You will need it to connect to the virtual machine.
 - iv. Inbound ports: open ports **22**, **443**, **9443**, and **8000** (all are required for StellarOne).
 - v. Disk: add a second data disk of at least **70 GB**.

4. Post-Deployment Configuration

- a. After the virtual machine has finished deploying, complete the following steps:
 - i. Ensure the 70 GB second disk is attached and recognized.
 1. Reboot the virtual machine if you added the second disk after initial deployment.
 - ii. Connect to the StellarOne server via SSH using the private key you downloaded during VM creation. Log in as user `txuser`.
 - iii. In the vShell command console, run the `oobe` command to change the default password.
 - iv. In the vShell command console, set the public IP address with the following commands:

```
env advertise-addr [YOUR VM PUBLIC IP]
service reload
```

- v. Confirm that inbound ports **8000** and **9443** are open for Stellar Agent communication.

StellarOne Onboarding to KVM

This section describes how to deploy StellarOne on to KVM (Kernel-based Virtual Machine).

Prerequisites

- The qcow2 packages provided by TXOne must be available and accessible to KVM 2.0.0
- The necessary networks have been properly created in KVM
- An extra disk (with more than 70GB of free disk space)
- The following server ports have been opened for StellarOne:

From	To	Open Port		Function
Stellar Agent	StellarOne	9443, 8000, 443	-	StellarOne's listening port for Stellar Agent
Stellar Agent (Windows-Legacy)	StellarOne	8000, 443	-	StellarOne's listening port for Stellar Agent (Windows-Legacy)
StellarOne	Stellar Agent	14336, 14337	-	Stellar Agent's listening port
StellarOne	Stellar Agent (Windows-Legacy)	14336, 14337	-	Stellar Agent (Windows-Legacy)'s listening port
StellarOne	License (PR) Server	443	https://odc.cs.txone-networks.com	For versions before StellarOne 3.1.3, StellarOne connects to global server port for license verification and renewal through HTTPS
StellarOne	License (PR) Server	443	https://apis.cs.txone.com https://odc.cs.txone-networks.com	For versions after StellarOne 3.1.3, StellarOne connects to

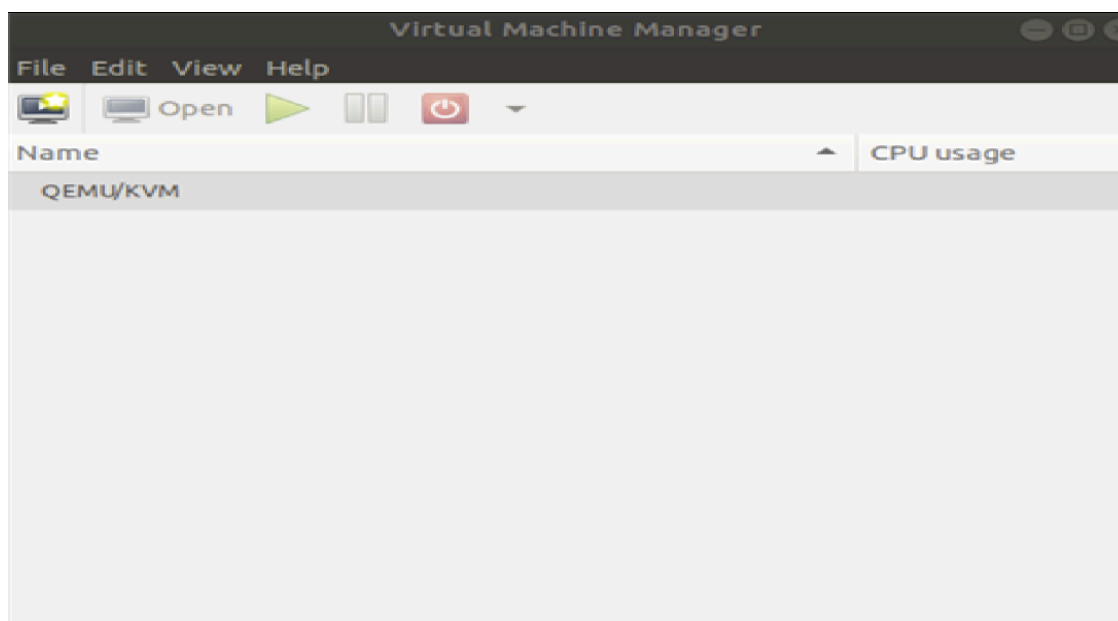
From	To	Open Port		Function
				global server port for license verification and renewal through HTTPS
Browser	StellarOne Web	443	-	StellarOne's listening port for web access through HTTPS
StellarOne	Active Update Server	443	https://txse-p.activeupdate.trendmicro.com/activeupdate https://txsp-p.activeupdate.trendmicro.com/activeupdate https://odc.cs.txone-networks.com	For StellarOne v2.0 and before, StellarOne connects to global server port for the Stellar Active Update through HTTPS
StellarOne	Active Update Server	443	https://ttauc.s.txone.com/protect https://ttauc.s.txone.com/enforce https://odc.cs.txone-networks.com	For StellarOne versions between v2.1 and V3.2, StellarOne connects to global server port for the Stellar Active Update through HTTPS
StellarOne	Active Update Server	443	https://ttauc.s.txone.com/protect https://ttauc.s.txone.com/enforce https://apis.cs.txone.com	For StellarOne V3.2 and after, StellarOne connects to global server port for the Stellar Active Update through HTTPS

From	To	Open Port		Function
			https://storage.cs.txone.com https://odc.cs.txone-networks.com	

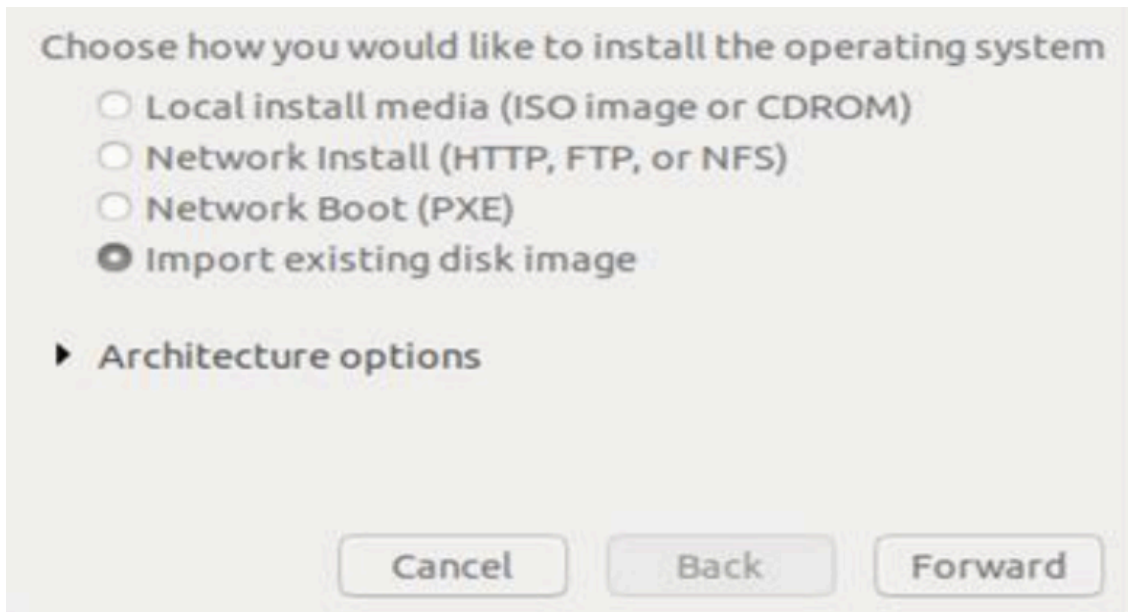
Deploying StellarOne on KVM

The below section details procedures of deploying StellarOne to KVM.

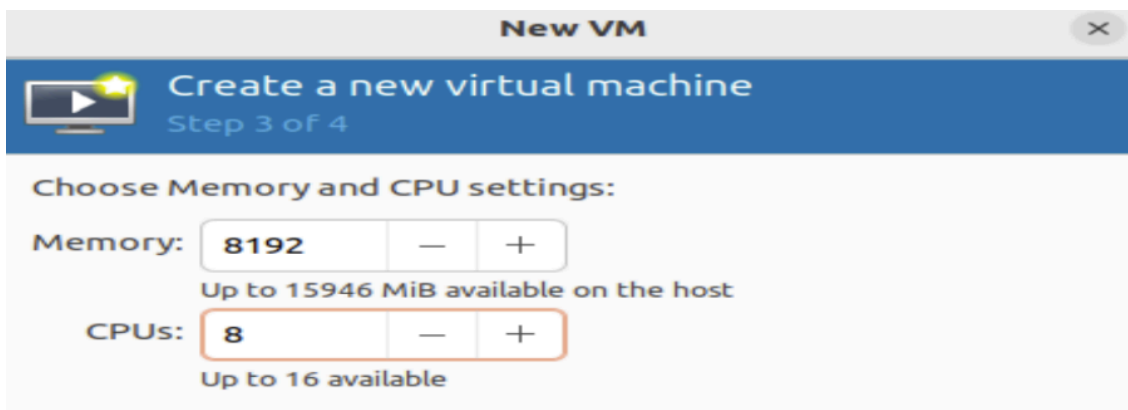
1. Open **Virtual Machine Manager**.
2. Click **Create a new virtual machine** icon in the top left corner.



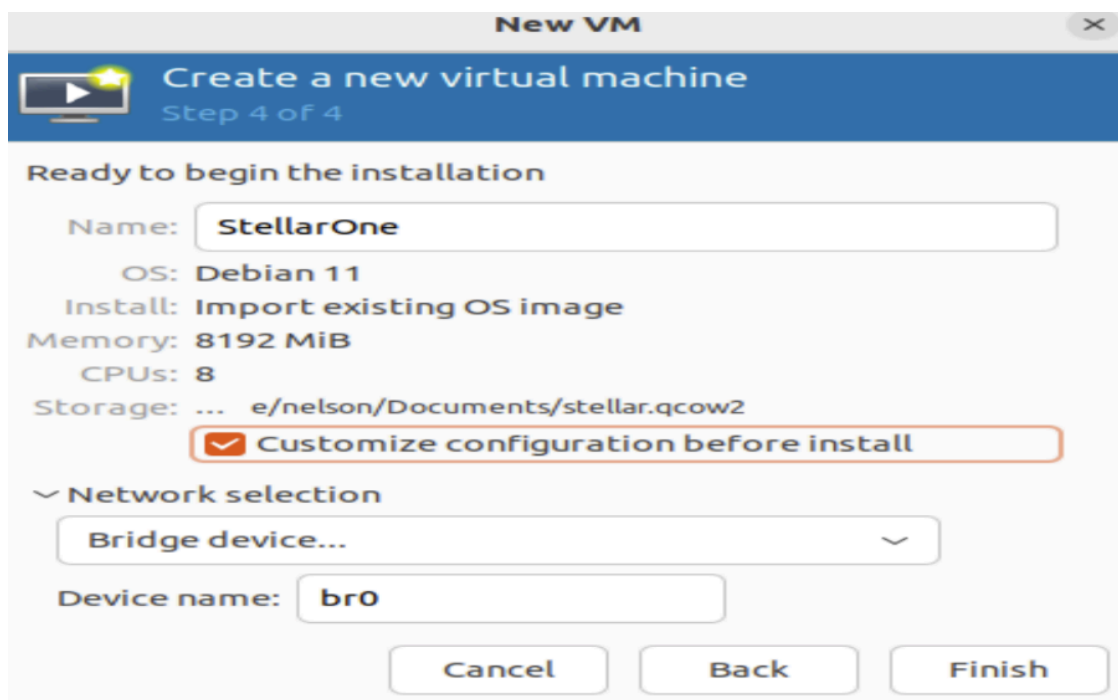
3. Select **Import existing disk image** and click **Forward**.



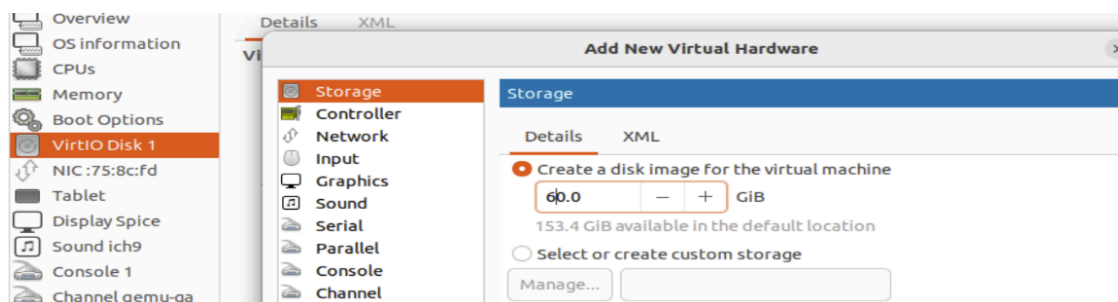
4. Click **Browse** and choose the **StellarOne** VM image.
5. From the operating system dropdown, select **Debian**.
6. Adjust memory allotment and number of CPU cores as per your requirements.



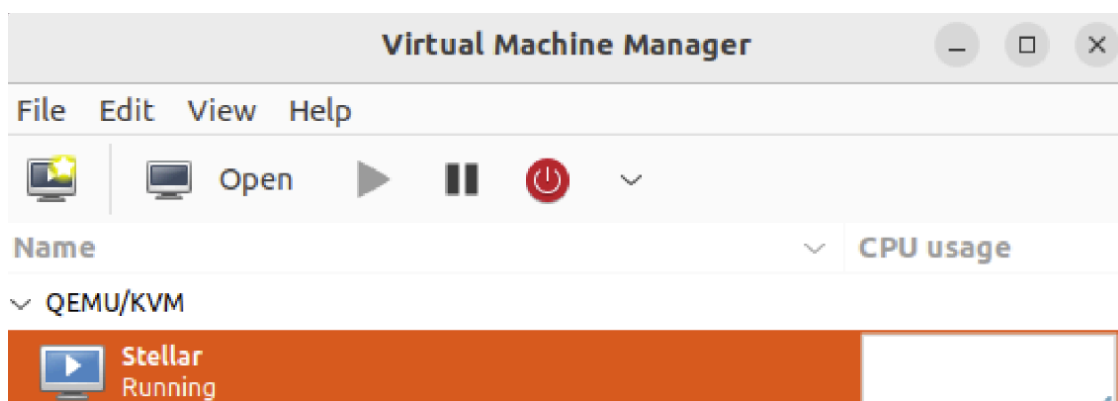
7. Input a virtual machine name, and check **Customize configuration before install**.



8. Ensure the **Network Selection** is set to **Bridge** device and click **Finish**.
9. Click **Add Hardware**, select **Storage**, and adjust the disk settings.



10. In the upper left corner, click **Begin Installation**.
11. StellarOne should now be running.



Accessing StellarOne CLI

The below section details procedures of accessing the **StellarOne CLI** after deployment to KVM.

1. Open the console of the **StellarOne** VM.
2. Login with the default credentials (refer to [StellarOne Default Login Account \(on page 32\)](#)).
3. Change the default password:
 - a. Enter "oobe".
 - b. Change the default password.
 - c. Re-login to the **StellarOne** CLI with your new password.
4. After logging in, you can enter "help" to see a list of available commands.

Getting the IP Address of the StellarOne Instance

The below section details the procedure for acquiring the IP address of your **StellarOne** instance.

1. Open the StellarOne CLI.
2. Enter "iface ls" to get the IP address of the StellarOne Instance.

Opening StellarOne Management Console

1. For VMware or Hyper-V system, launch the StellarOne instance and then run the OOB process in the setup CLI.

- a. Use the default credentials when logging on for the first time:

- Username: txuser

- Password: txone

- b. Change the default credentials to your desired username and password.



Important:

To enhance security, it is recommended to create a safe username and strong password.



Note:

For StellarOne instance deployed from AMI on AWS EC2 platform, or on the Microsoft Azure platform, the OOB process is not required.

2. Open a web browser and enter the StellarOne address in the following format: `https://<targetserver IP address>`. The logon screen appears.
3. Enter your credentials (user ID and password).

Use the default credentials of administrator when logging on for the first time:

- User ID: admin
 - Password: txone
4. Click **Log On**.
5. If this is the first time the StellarOne instance being logged on, follow procedures below to complete the initial settings.

- a. The **Login Information Setup** window appears and prompts you to change password. Confirm your password settings by:

- specifying your new password in the **New Password** text field.
- specifying the password again in the **Confirm Password** text field.

**Note:**

- For StellarOne 1.2 or above, the default login name is always `admin` and can not be changed by the user.
 - For StellarOne 1.0/1.1, in addition to changing the password, the user is also required to change the default login name in this step. The new login name cannot be `admin`, `administrator`, `auditor` or `root`.
-

- b. Click **Confirm**. You will be automatically logged out. The **Log On** screen will appear again.
- c. Log on again using your new credentials. The **License Activation** window appears.
- d. StellarOne recognizes two license formats (**License Key** and **Activation Code**) available from different sales channels. See [Comparison of License Formats \(on page 69\)](#) first to find the license format that matches the given license data.
- e. See the following instructions for how to activate license depending on the network environment.

- **StellarOne has Internet connection**

- i. Click **License Key**.
- ii. Specify the License Key or Activation Code in the text field.

- **StellarOne has NO Internet connection**

- If the given license format is Activation Code, click **License Key** and specify the Activation Code in the text field.
- If the given license format is License Key, use it to download the License File (a .txt file). See [Getting the License File \(on page 69\)](#) for the detailed procedures. After getting the License File, click **License File** and import it.

- f. Click **Apply**.

- g. A success message appears. The license information also appears at the bottom of the **License Activation** window. Check if it matches the given license data.



Note:

See [Resolving Licensing Issues \(on page 96\)](#) if licensing related error messages appear.



Note:

The **License Activation** window accepts the StellarOne server's primary license only. If you intend to use a StellarDiscover add-on license, first activate the StellarICS primary license here, then apply the StellarDiscover add-on license from **Administration > License** after you complete the initial setup.

- h. Click **Continue**.
- i. The **End User License Agreement and TXOne OT Intelligent Trust** window appears. Click the links to read the documents carefully and click the checkboxes to proceed to next step.



Note:

It is recommended to enable **TXOne OT Intelligent Trust** to enhance security deployment. See [OT Intelligent Trust \(on page 70\)](#) for more details.

- j. Specify the time settings such as the **Date and Time** as well as the **Time Zone**, and then click **Continue**.
- k. The StellarOne console is ready for use now.



Note:

After the initial settings are completed, the StellarOne allows various user accounts to log on remotely via a web browser.

6. (Optional) You can change your password by clicking the ID icon at the top right corner of the screen, and then selecting **Change Password**.
7. (Optional) For security reasons, you can manually log off by clicking the ID icon at the top right corner of the screen.
- a. A pop-up **Log Off** window appears. Click **Yes** to log out of StellarOne.



Note:

You will be automatically logged off the console if no operations are performed within 30 minutes.

Comparison of License Formats

StellarOne recognizes two formats of product license purchased from TXOne Networks or from a TXOne Networks authorized reseller.

Table 12. Comparison of Two Different License Formats

		License Key	Activation Code
Length		19 characters	37 characters
Example		FIJN-HPYB-XXXX-XXXX	TE-24RF-Q9UN9-S9QQN-XXX XX-XXXXXX-XXXXXX
New license	Online	√	√
	Offline	License File	√
License renewal	Online	√	√
	Offline	License File	N/A



Note:

A License Key is required for downloading a License File.

Getting the License File

If the given license format is License Key, you can use it to download the license file, which can be used to activate a new license or renew the current license when StellarOne has no Internet connection. To get a license file, follow procedures below.

1. Go to **Administration > License**
2. Click the **New License Key/File** button.
3. The **New License** window appears.
4. When you log on StellarOne using your new credentials after the **Login Information Setup** procedure, the **License Activation** window appears.
5. Click **License File**.
6. Click **Copy License File Download Link** at the bottom of the **New License** window.



Important:



A License Key is required for downloading a License File. An Activation Code cannot be used to download a License File. See [Comparison of License Formats \(on page 69\)](#) for more details.

7. **The Download Link has been copied** toast message appears.
8. Open the copied link in a web browser on a computer with Internet connection.
9. You will be directed to the TXOne **License File Management** screen. Specify the given License Key in the **License Key** field, and then click **Download**.

Figure 40. TXOne License File Management

License File

Product Serial Number

a8fc1f7e-7788-11ed-8b70-0050569a90f7

License Key

Download

10. A pop-up window appears showing the license information. Read it carefully and click **Yes** for downloading the license file.



Tip:

You can send the license file to StellarOne that has no Internet connection via the internal secured network or a trusted portable device.

OT Intelligent Trust

When enabled, TXOne OT Intelligent Trust shares anonymous threat information with the Smart Protection Network, allowing TXOne to rapidly identify and address new threats. You can disable TXOne OT Intelligent Trust anytime on this console.

1. Go to **Administration > OT Intelligent Trust**.
2. Click the **Learn More** for visiting TXOne's OT threat research website.
3. To enable TXOne OT Intelligent Trust, toggle on the **Enable TXOne OT Intelligent Trust (recommended)**.

Chapter 4. Configuring StellarOne via Command Line Interface (CLI)

This chapter describes how to configure some settings for StellarOne via command line interface (CLI).

Topics in this chapter include:

- [Using the StellarOne Command Line Interface \(CLI\) \(on page 72\)](#)
- [Configuring the IP Address via CLI \(on page 72\)](#)
 - [Configuring the Advertise Address via CLI \(on page 74\)](#)
- [Modifying Communication Ports via CLI \(on page 76\)](#)
- [Managing Docker Network via CLI \(on page 78\)](#)
- [Setting Up ACL \(Access Control List\) via CLI \(on page 78\)](#)
- [Changing Language Settings via CLI \(on page 80\)](#)
- [Resetting Administrator's Password via CLI \(on page 81\)](#)

Using the StellarOne Command Line Interface (CLI)

The following section describes how to log on StellarOne and get a list of available commands via command line interface (CLI).

1. Open the StellarOne VM console.
2. Log on by entering the username and password changed during the OOB process. See [Opening StellarOne Management Console \(on page 66\)](#) for more details.



Note:

For StellarOne instance deployed from AML on AWS EC2 platform, and on the Microsoft Azure platform, the OOB process is not required. Enter admin to log on for AWS EC2 and txuser for Microsoft Azure.

3. After logging on the StellarOne console, type `help` command for a list of available commands.

Configuring the IP Address via CLI

The following section describes procedures of configuring the IP address settings for StellarOne instance via CLI.

1. Type `iface ls` to get the IP address of the StellarOne instance.

Figure 41. Getting the IP Address of StellarOne

```

$ iface ls
[
  {
    "Name": "lo",
    "Family": "inet",
    "Method": "loopback"
  },
  {
    "Name": "eth0",
    "Family": "inet",
    "Method": "dhcp"
  }
]
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 odisc noqueue state UNKNOWN group default qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 odisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:fc:65:af brd ff:ff:ff:ff:ff:ff
    inet 192.168.68.147/24 brd 192.168.68.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe65:af64 scope link
        valid_lft forever preferred_lft forever
$
  
```

2. Type `iface update` command for updating the settings of current network interface. For example, the following command sets the interface `eth0` to a static IP address `10.7.19.157/24` with the Gateway IP address `10.7.19.254`.

```
iface update eth0 --method static --address 10.7.19.157 --netmask 255.255.255.0 --gateway 10.7.19.254
```

3. Check if the network interface settings are correct, and then type the following command to execute the change.

```
iface restart eth0
```

4. Type following command again for viewing the new network interface settings.

```
iface ls
```

Figure 42. Viewing New Network Settings

```
[
  {
    "Name": "lo",
    "Family": "inet",
    "Method": "loopback"
  },
  {
    "Name": "eth0",
    "Family": "inet",
    "Method": "static",
    "Address": "10.7.19.157",
    "Netmask": "255.255.255.0",
    "Gateway": "10.7.19.254"
  }
]

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:2f:05:2d brd ff:ff:ff:ff:ff:ff
    inet 10.7.19.157/24 brd 10.7.19.255 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe2f:52d/64 scope link
        valid_lft forever preferred_lft forever
```

5. Use the `resolv add` command to add a DNS server and `resolv ls` to view the DNS server list. For example, the following command adds 8.8.8.8 to the DNS server list

```
resolv mode custom
```

```
resolv add 8.8.8.8
```

6. Type following command to view the DNS server settings

```
resolv ls
```

Figure 43. Viewing DNS Server Settings

```
$ resolv mode custom
$ resolv add 8.8.8.8
8.8.8.8 is added
$ resolv ls
Custom Mode
8.8.8.8
```

7. Type following command to reboot the VM.

```
reboot
```

**Note:**

It is suggested to configure the StellarOne advertise address for the agents to communicate with if there's possibility to change StellarOne IP address in the future. See [Configuring the Advertise Address via CLI \(on page 74\)](#) for more details.

Configuring the Advertise Address via CLI

The following section describes how to configure the IP address or FQDN as the StellarOne advertise address via CLI.

1. Type `help` command for a list of available commands.
2. Type `env` and find the `advertise-addr` command.

Figure 44. The `advertise-addr` command

```
$ help
vShell, version ab2e3bc
The commands provided in:
access-list  Manage the IP whitelists
dx           Connection test for target server
env         Manage system environment variables
exit        Exit this shell
help        List all command usage
iface       Manage the network interfaces
ping        Test the reachability of a host
poweroff    Shut down the machine immediately
pwd         Change the root user password
reboot      Restart the machine immediately
resolve     Manage the domain name server
scp         Send files via scp
ssh         SSH to a device
service     Manage the StellarOne service
sftp        Send files via sftp
web         Commands of the web management console
stellar     Commands of the Stellar products
locale      Locale setting
network     Manage network for the StellarOne service

Shortcut table:
Tab         Auto-complete or switch among options available
Ctrl + A    Go to the head of the line (Home)
Ctrl + E    Go to the tail of the line (End)
Ctrl + D    Delete the character located at the cursor
Ctrl + L    Clear the screen

$ env
ls          List the OS and Service environment variables
hostname    Edit the variable in /etc/hostname
advertise-addr Set the IP address or FQDN to advertise, or type "default" to use the default IP of the host
logseverity Set the debug log level [default info verbose]
```

3. Specify the advertise address for StellarOne after the `advertise-addr` command. The following example uses the FQDN as the advertise address:

```
env advertise-addr S1.txone.com
```

4. (Optional) If the specified advertise address can not be resolved, type the `--force` command after it to force the setup:

```
env advertise-addr S1.txone.com --force
```

**Note:**



In the examples above, S1.txone.com is used, but this should be replaced. You can choose to specify the IP address, FQDN, or type `default` to use the default IP address for StellarOne.

5. Reload the StellarOne web console by typing:

```
service reload
```

```

$ env advertise-addr Sl.txone.com
vShell: unable to resolve address Sl.txone.com or it's an invalid IP. Try to add --force after the address to force the setup.
$ env advertise-addr Sl.txone.com --force
Successfully set up advertise address: Sl.txone.com. Please reload the service to take effect.
$ service reload
Start to reload services....

```

6. Type the following command to check the advertise address settings.

env ls

Figure 45. Checking Advertise Address Settings

```
$ env ls
Hostname: ODC
Status: RUNNING
Product Serial Number: b2
Version: 2.2.1148
Advertise Address: Sl.txone.com
DPI Engine Version:
DPI Pattern Version:
StellarProtect (Legacy Mode) Agent Up Port:8000
StellarProtect (Legacy Mode) Agent Down Port:14336
StellarProtect Agent Up Port: 9443
StellarProtect Agent Down Port:14336
Locale: en
$
```

7. After the setup, the agent installer package and the SAML SSO (Single Sign-On) metadata file downloaded from StellarOne should contain the configured advertise address, allowing the quick deployment for the agents to communicate with StellarOne or SSO login via the advertise address.

- See the examples below as the agent's config files (StellarSetup.ini) that contain the configured advertise address.

Figure 46. Use IP address as the advertise address in StellarSetup.ini

```
[shared_license]
product_serial_number = '████████████████████████████████████████'
txone_license_file =
txone_license_env = prod

[shared_server]
host = 10.8.150.84
cert = server.crt
```

Figure 47. Use FQDN as the advertise address in StellarSetup.ini

```
[shared_license]
product_serial_number = 7████████████████████████████████████████
txone_license_file =
txone_license_env = prod

[shared_server]
host = S1.txone.com
cert = server.crt
```

- See the example below as the SAML SSO metadata files that contain the configured advertise address.

Figure 48. Use IP address or FQDN as the advertise address in SAML SSO metadata file



Modifying Communication Ports via CLI

Below section describes how to modify the communication ports for StellarOne instance via CLI.

1. Type `env ls` command for the list of current communication ports.

Figure 49. List of Current Communication Ports

```
$ env ls
Hostname: ODC
Status: RUNNING
Product Serial Number: 7e [bar] a9
Version: 2.2.1148
Advertise Address: Not Set
DPI Engine Version:
DPI Pattern Version:
StellarProtect (Legacy Mode) Agent Up Port:8000
StellarProtect (Legacy Mode) Agent Down Port:14336
StellarProtect Agent Up Port: 9443
StellarProtect Agent Down Port:14336
Locale: en
```

2. Type `stellar` command for available agents to appear for selection.

Figure 50. Available Agents for Selection

```
$ stellar
set-enforce-ports Edit the communication ports for Stellar Enforce agents
set-protect-ports Edit the communication ports for Stellar Protect agents
```

3. Select one of the agents to edit its communication port.

Figure 51. Select the Agent for Editing Communication Port

```
$ stellar set-enforce-ports
set-enforce-ports Edit the communication ports for Stellar Enforce agents
set-protect-ports Edit the communication ports for Stellar Protect agents.
```

4. Input the valid value for <up-port> and <down-port>.

- <up-port>: Port for receiving data from agents.
- <down-port>: Port for sending command to agents



Note: Make sure not to use StellarOne's service port. Please refer to **Table 2-7. StellarOne Occupied Ports** in [Outbound Ports and FQDN Used \(on page 23\)](#).

Figure 52. Agent's Communication Ports

```
$ stellar set-enforce-ports 8000 14336
Port for receiving data from Stellar Enforce agents: 8000
Port to send commands to Stellar Enforce agents: 14336

Successfully set up ports for Stellar Enforce.
Please reload services to take effect.
```

5. Reboot.

**Important:**

Please note the previously installed package does not contain the new port setting. Be sure to do either of the following actions after changing the communication ports for StellarOne via CLI.

- Download the agent's installer package containing the new port setting from StellarOne, and install it on the agent.
- Modify the port setting accordingly in the `StellarSetup.ini` file in the agent's existing installer package, and reinstall it on the agent.

Managing Docker Network via CLI

The following section describes how to manage docker network on vShell for StellarOne via CLI.

1. If `169.254.0.0/16` IP range is used in your network setting, please type `network internal-service-update <New IP>` command to set a new IP address for converting IP/16 subnet mask for docker daemon.
2. If you want to restore docker daemon back to the default-address-pools (`169.254.0.0/16`), type `network internal-service-reset` command.
3. Type `network internal-service-list` command to display the address pools of docker daemon configuration.

Setting Up ACL (Access Control List) via CLI

The vshell access-list command allows you to restrict network access to the service by two trust list.

- SSH: When enabled, only the IP addresses on this list can access the SSH service (tcp:22) in the system.
- Web: When enabled, only the IP addresses on this list can access the dashboard service.

Querying the Status

To obtain the active statuses, port numbers and IPs/CIDRs on the trust list.

```
$ access-list ls
```

```
SSH(tcp:22)
```

```
Status: Disabled
```

```
Web(tcp:443)
```

```
Status: Disabled
```

Adding Clients to the Trust List

You can add client IPs or Classless Inter-Domain Routing (CIDR).

```
$ access-list append SSH 1.1.1.1
added! Please check the trust list.
$ access-list append SSH 1.1.1.0/24
1.1.1.0/24 added Please check the trust list.
```

Deleting Clients from the Trust List

You can delete client IPs or Classless Inter-Domain Routing (CIDR).

```
$ access-list trim SSH 1.1.1.1
removed! Please check the trust list.
$ access-list removed SSH 1.1.1.0/24
1.1.1.0/24 removed Please check the trust list.
```

Enabling/Disabling the ACL of Modules



Note:

1. If you log in over SSH, enabling the SSH ACL will immediately force you out of your SSH session.
2. Before you enable the ACL, please add clients to the trust list. If clients are not added before ACL is enabled, all clients will be blocked from connecting. If clients are not added to the trust list before ACL is enabled, it will be necessary for the administrator to edit the trust list directly.
3. When the ACL is enabled, without any defined rules, the device blocks all network traffic.

```
$ access-list up SSH
SSH enabled! Please check the trust list.
$ access-list down SSH
SSH disabled! Please check the trust list.
```

Shortcut Table

Table 13. Shortcut Table

Keyboard Shortcut	Function
Tab	Auto-complete or choose the next suggestion on the list
Ctrl+A	Go to the head of the line (Home)
Ctrl+E	Go to the end of the line (End)

Table 13. Shortcut Table (continued)

Keyboard Shortcut	Function
Ctrl+D	Delete the character located at the cursor
Ctrl+L	Clear the screen

Changing Language Settings via CLI

Below section describes how to change language settings for StellarOne via CLI. The default language for StellarOne web console is English. You can change the language to Japanese following below procedures.

1. Type `locale ja` command to switch the language to Japanese.
2. Reload the StellarOne web console

Figure 53. Reload StellarOne console

```
$ help
vShell, version ab2e3bc
The commands provided in:
  access-list  Manage the IP whitelists
  dx           Connection test for target server
  env          Manage system environment variables
  exit         Exit this shell
  help        List all command usage
  iface       Manage the network interfaces
  ping        Test the reachability of a host
  poweroff    Shut down the machine immediately
  pwd         Change the root user password
  reboot      Restart the machine immediately
  resolv      Manage the domain name server
  scp         Send files via scp
  ssh         SSH to a device
  service     Manage the StellarOne service
  sftp        Send files via sftp
  web         Commands of the web management console
  stellar     Commands of the Stellar products
  locale      Locale setting
  network     Manage network for the StellarOne service

Shortcut table:
  Tab         Auto-complete or switch among options available
  Ctrl + A    Go to the head of the line (Home)
  Ctrl + E    Go to the tail of the line (End)
  Ctrl + D    Delete the character located at the cursor
  Ctrl + L    Clear the screen

$ locale ja
Successfully language setting for locale.
Please reload StellarOne console to take effect.
```

3. Type `env ls` command to check current language settings.

Figure 54. Check Language Settings

```
$ env ls
Hostname: ODC
Status: RUNNING
Product Serial Number: 7a...a9
Version: 2.2.1148
Advertise Address: Not Set
DPI Engine Version: 2.0.11.33e2e1+turbo
DPI Pattern Version: SDP_230228_08
StellarProtect (Legacy Mode) Agent Up Port:8000
StellarProtect (Legacy Mode) Agent Down Port:14336
StellarProtect Agent Up Port: 9443
StellarProtect Agent Down Port:14336
Locale: ja
```

Resetting Administrator's Password via CLI

The following section describes how to reset administrator's password for StellarOne via CLI.

1. Type `web reset admin` command to reset administrator's password.
 2. The `reset OK!` message appears. The administrator's password has been reset.
 3. Use the default credentials (user ID: `admin` / password: `txone`) to log on the StellarOne web console.
-



Note:

For StellarOne 1.0/1.1, the default login name (user ID) is required to be changed by users. Be sure to use the changed default login name for accessing StellarOne 1.0 or 1.1.

4. The **Login Information Setup** window appears and prompts you to change password. Confirm your password settings by:
 - a. specifying your new password in the **New Password** text field.
 - b. specifying the password again in the **Confirm Password** text field.
5. Click **Confirm**. You will be automatically logged out. The **Log On** screen will appear again.
6. Log on again using your new credentials.

Chapter 5. Upgrade

This chapter describes the supported upgrade paths and methods for TXOne StellarOne.



Important:

Since some requirements such as the sizing table or ports and FQDN used by StellarOne may differ in different versions, it is recommended to check [Installation Planning \(on page 17\)](#) before performing the upgrade.

Topics in this chapter include:

- [Supported Upgrade Paths \(on page 84\)](#)
- [Upgrade Methods \(on page 86\)](#)

Supported Upgrade Paths

The following table illustrates the supported upgrade paths for StellarOne installed in VMware or Windows Hyper-V system.

Table 14. Supported Upgrade Paths

Platform	Current Version	Supported Target Upgrade Version	Firmware Upgrade*	Mount Upgrade
VMWare Hyper-V	3.3	5.0	√	√
	3.2 Service Pack 1 Patch 2	3.3	√	√
	3.2 Service Pack 1 Patch 1	3.2 Service Pack 1 Patch 2	√	√
	3.2	3.2 Service Pack 1 / 3.2 Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2	√	√
	3.1 Patch 2	3.2 / 3.2 Service Pack 1 / 3.2 Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2	√	√
	3.1 Patch 1	3.1 Patch 2 / 3.2 / 3.2 Service Pack 1 / 3.2 Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2	√	√
	3.1	3.1 Patch 1 / 3.1 Patch 2 / 3.2 / 3.2 Service Pack 1 / 3.2 Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2	√	√
	3.0 Service Pack 1	3.1 / 3.1 Patch 1 / 3.1 Patch 2 / 3.2 / 3.2 Service Pack 1 / 3.2	√	√

Table 14. Supported Upgrade Paths (continued)

Platform	Current Version	Supported Target Upgrade Version	Firmware Upgrade*	Mount Upgrade
		Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2		
	3.0	3.0 Service Pack 1 / 3.1 / 3.1 Patch 1 / 3.1 Patch 2 / 3.2 / 3.2 Service Pack 1 / 3.2 Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2	√	√
	2.2	3.0 / 3.0 Service Pack 1 / 3.1 / 3.1 Patch 1 / 3.1 Patch 2 / 3.2 / 3.2 Service Pack 1 / 3.2 Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2	√	√
	2.1	2.2 / 3.0 / 3.0 Service Pack 1 / 3.1 / 3.1 Patch 1 / 3.1 Patch 2 / 3.2 / 3.2 Service Pack 1 / 3.2 Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2	√	√
	2.0	2.1 / 2.2 / 3.0 / 3.0 Service Pack 1 / 3.1 / 3.1 Patch 1 / 3.1 Patch 2 / 3.2 / 3.2 Service Pack 1 / 3.2 Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2	√	√

Table 14. Supported Upgrade Paths (continued)

Platform	Current Version	Supported Target Upgrade Version	Firmware Upgrade*	Mount Upgrade
	1.2 Patch 1 (1.2.2114)	2.0 / 2.1 / 2.2 / 3.0 / 3.0 Service Pack 1 / 3.1 / 3.1 Patch 1 / 3.1 Patch 2 / 3.2 / 3.2 Service Pack 1 / 3.2 Service Pack 1 Patch 1 / 3.2 Service Pack 1 Patch 2	N/A	√
	1.2	2.0 / 2.1	N/A	√
	1.2	1.2 Patch 1 (1.2.2114)	√	√
VMWare	1.1	1.2 / 1.2 Patch 1 (1.2.2114)	√	√
	1.0	1.1 / 1.2 / 1.2 Patch 1 (1.2.2114)	N/A	√

**Important:**

- The 2nd disk space requirement for StellarOne varies depending on whether certain feature is enabled. See [Hardware Requirements for Deploying StellarOne on VMware/Hyper-V/AWS EC2 \(on page 19\)](#) to check if the storage requirement is fulfilled.
- TXOne Networks recommends always using firmware upgrade to update StellarOne if both upgrade options are available.

Upgrade Methods

This section describes two methods to upgrade StellarOne installed in VMware or Windows Hyper-V system.

**Note:**



Since Stellar Agent (Windows) and Stellar Agent (Windows-Legacy) agent events will be combined into one display page after upgrading StellarOne to version 3.1, some events may exceed the limit and thus be deleted. Ensure you check the current event quantities and adjust the **Log Purge** configuration before the upgrade. Refer to the *StellarOne Administrator Guide* for more information about how to adjust the **Log Purge** settings.

**Note:**

Discover agents and seat usage after upgrading. If you are upgrading from an earlier StellarOne 5.0 build to a build that supports StellarDiscover add-on licenses, Discover agents that were already registered continue to consume StellarICS seats after the upgrade. After you apply a StellarDiscover add-on license, those agents are automatically reassigned to the Discover add-on seat pool and no longer consume StellarICS seats. For instructions on applying a StellarDiscover add-on license, see the *StellarOne Administrator Guide*.

Topics in this section include:

- [Firmware Upgrade \(on page 87\)](#)
- [Mount Upgrade \(VMware\) \(on page 88\)](#)
- [Mount Upgrade \(Hyper-V\) \(on page 90\)](#)

Firmware Upgrade

This section describes how to perform firmware upgrade via the StellarOne web console.

1. Download the .acf upgrade patch file (e.g., TX0ne-S1-acus.fw-3.x.xxxx.acf) from [MyTxOne](#).
2. Log on the StellarOne web console and go to **Administration > Firmware**.
3. Click **Upload ACF File** and select the .acf file downloaded in *Step 1*, and then click **Apply**.
4. Wait until the following window appears and read the upgrade notice carefully.

Figure 55. Firmware Install

**Note:**

Before executing the firmware upgrade, please create a back up of the VM files first.

5. Click **Install Now** to start the upgrade.

Mount Upgrade (VMware)

This section describes how to perform mount upgrade for StellarOne on VMware ESXi system. The mount upgrade is performed by attaching the external disk of previous StellarOne instance to the StellarOne instance running new firmware version. The previously configured settings will be transferred to the new StellarOne instance, including:

- The UUID
- The pattern and firmware
- The agent list, policy settings, and StellarOne certificates
- The system configuration, including license, account information, security policies, and proxy/SSO settings
- Security event logs

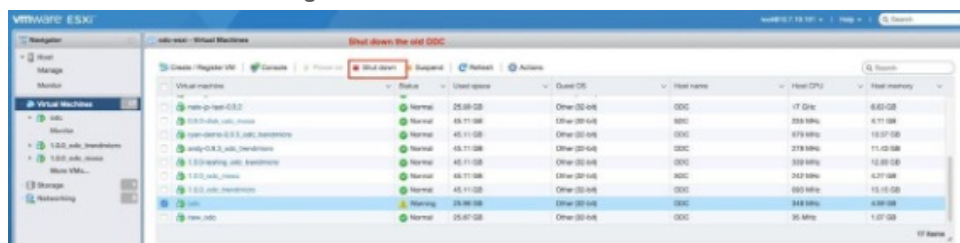


Important:

- Before executing a mount upgrade, please create a backup of the VM files first.
- StellarOne 2.0 ONLY supports mount upgrade from version 1.2 or 1.2 Patch 1. Make sure you upgrade StellarOne 1.1 to 1.2 or 1.2 Patch 1 before upgrading to 2.0.
- StellarOne 1.1 ONLY supports mount upgrade from version 1.0.

1. Deploy a new StellarOne instance. See [Deploying StellarOne on the VMware ESXi \(on page 32\)](#) for detailed instructions.
2. Close the previous StellarOne instance.

Figure 56. Shut Down the Previous VM



3. Attach the external disk of the previous StellarOne instance to the new StellarOne instance.

Figure 57. Select Existing (Previous) Hard Disk

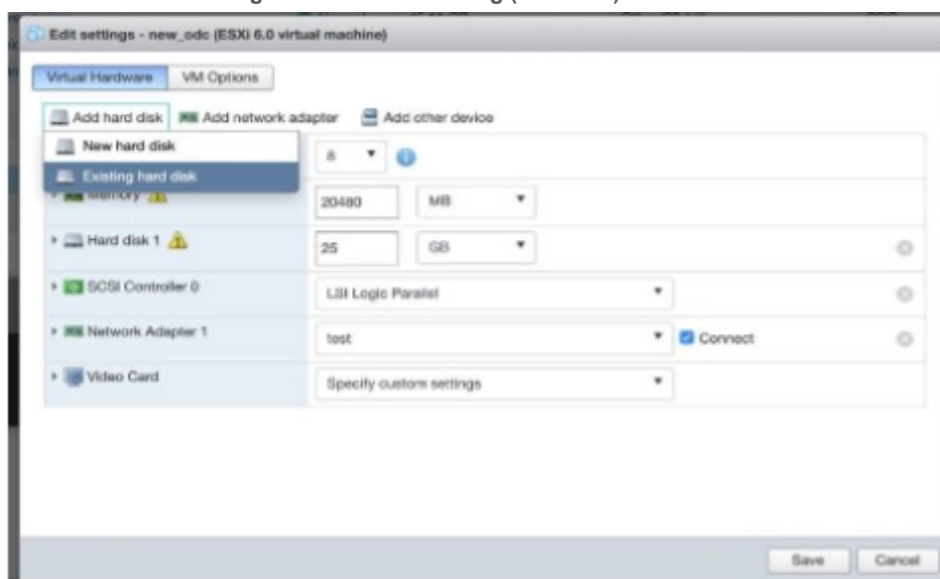
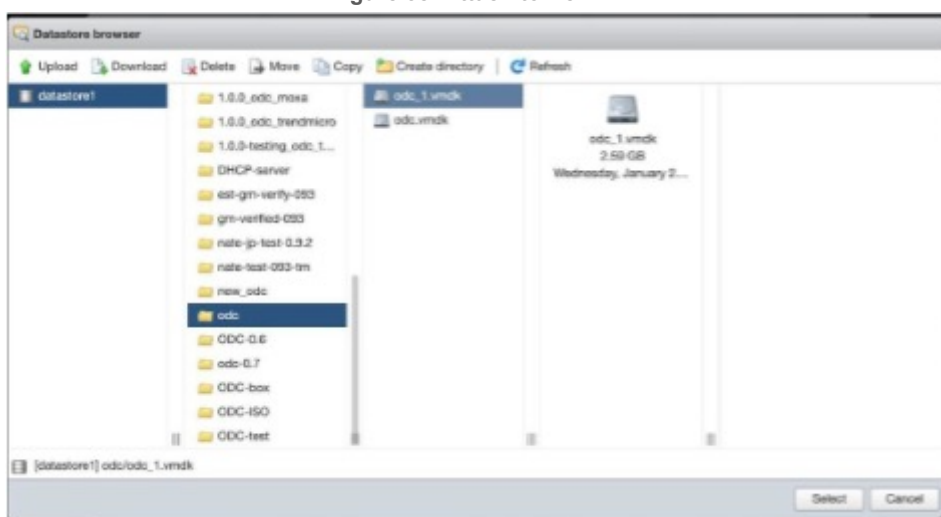


Figure 58. Attach to New VM



4. Boot up the new StellarOne instance.

**Note:**

The data of the previous StellarOne instance has been migrated to the new StellarOne instance.

5. The IP address of the new StellarOne instance must be the same as that of the previous StellarOne instance. If not, manually configure the IP address so the new StellarOne instance and agents can be connected to each other. Next time when the agents synchronize their status with the server, they will connect to the new StellarOne. By default, the agents synchronize with the server every 20 minutes.
6. If you want to change the language setting to Japanese for the new StellarOne instance, see [Changing Language Settings via CLI \(on page 80\)](#).

Mount Upgrade (Hyper-V)

This section describes how to perform mount upgrade for StellarOne in Windows Hyper-V system. The mount upgrade is performed by attaching the external disk of previous StellarOne instance to the StellarOne instance running new firmware version. The previously configured settings will be transferred to the new StellarOne instance, including:

- The UUID
- The pattern and firmware
- The agent list, policy settings, and StellarOne certificates
- The system configuration, including license, account information, security policies, and proxy/SSO settings
- Security event logs

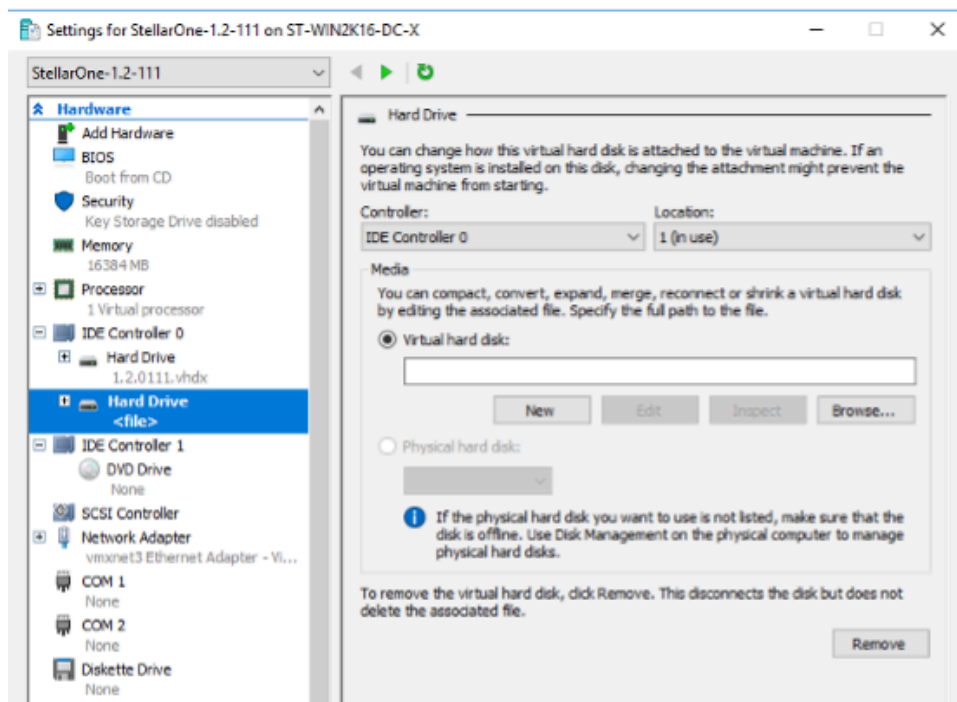


Important:

- Before executing a mount upgrade, please create a backup of the VM files first.
- StellarOne 2.0 ONLY supports mount upgrade from version 1.2 or 1.2 Patch 1.

-
1. Deploy a new StellarOne instance. See [Deploying StellarOne to a Hyper-V System \(on page 38\)](#) for deployment details.
 2. Close the previous StellarOne instance.
 3. Click **Browse** and choose the existing disk.
 4. Attach the external disk of previous StellarOne to the new StellarOne instance.

Figure 59. Shut Down the Previous VM



5. Boot up the new StellarOne instance.

**Note:**

The data of the previous StellarOne instance has been migrated to the new StellarOne instance.

6. The IP address of the new StellarOne instance must be the same as that of the previous StellarOne instance. If not, manually configure the IP address so the new StellarOne instance and agents can be connected to each other. Next time when the agents synchronize their status with the server, they will connect to the new StellarOne. By default, the agents synchronize with the server every 20 minutes.
7. If you want to change the language setting to Japanese for the new StellarOne instance, see [Changing Language Settings via CLI \(on page 80\)](#).

Chapter 6. Getting Help and Troubleshooting

Learn about the following topics:

- [Troubleshooting Resources \(on page 94\)](#)
- [Contacting TXOne Networks \(on page 100\)](#)
- [Other Resources \(on page 101\)](#)

Troubleshooting Resources

Before contacting technical support, consider visiting the following TXOne Networks online resources.

Self-Diagnosis

Identify and troubleshoot low disk space issues with StellarOne.

- [Resolving Low Disk Space Issues \(on page 94\)](#)
- [Resolving Licensing Issues \(on page 96\)](#)

Resolving Low Disk Space Issues

To keep you informed of your disk usage, and any potential issues that may arise, StellarOne displays banners or modal windows at certain disk usage thresholds. When encountering such errors, you can check the following table for the features affected and take actions as suggested in the **Workaround** column.

However, the most efficient way to get rid of low disk space error caused by the 2nd HDD is to add more space to the drive or partition. Follow the procedures:

1. Go to **Dashboard** to check **Widgets for Monitoring Disk Usage** section in *StellarOne Administrator Guide* for the current disk usage.
2. Go to **Dashboard** to check the Disk Usage widgets.
3. Check the following table for the features that will become unavailable at certain usage thresholds.
4. Extend the drive or partition which is running out of space.



Note:

- If you take snapshots for the StellarOne virtual machine, ensure the snapshot files be consolidated before extending the partition space.
 - Contact your support provider if you need to fix the low disk space error caused by the 1st HDD.
-

Table 15. Unavailable Features List due to Low Disk Space

Unavailable Features	Menu Path	Description	Usage Threshold	Workaround
Firmware upgrade	Administration > Update > Firmware	Unable to import the StellarOne firmware for upgrade	• 1st HDD disk free space less than 5 GB + the size of customer-specific data, such as 2nd HDD's other partitions, the number of uploaded agent patches and collected agent logs. • 2nd HDD free disk space: ◦ System partition: less than 5 GB ◦ Database partition: less than 30%	Free up the 2nd HDD disk space on System partition by: • Deleting unused patch files • Purging unwanted agent debug logs

Table 15. Unavailable Features List due to Low Disk Space (continued)

Unavailable Features	Menu Path	Description	Usage Threshold	Workaround
Agent patch file import	Administration > Downloads/Updates > Agent StellarOne debug page (for SEG purpose)	Unable to import Agent patch files	• 2nd HDD disk usage on System partition less than 20% • 2nd HDD disk usage on Database partition less than 20%	
Agent debug tool collection		Unable to collect the debug logs for Stellar Agent (Windows) or Stellar Agent (Windows-Legacy)		
Debug log level selection		Unable to select the Info or Verbose log level. Only the Warn (Default) level logs can be collected.		
Server debug tool collection		Unable to collect the debug logs for the StellarOne server		

Resolving Licensing Issues

The following table provides more information about some licensing error messages and possible actions to take when issues occur during license activation or renewal.

Table 16. Licensing Error Messages and Suggested Actions

Error Message	Description	Possible Actions
Unable to convert the license from full to trial version. To continue using the product, please contact your sales representative for renewing the license.	Once upgraded to a full license, the full license cannot be converted to a trial license.	• Ensure you enter the valid full license data. • Contact your sales representative for renewing the full license.
The product license is not supported.	The license you supplied cannot be applied as a StellarDiscover add-on license. This message appears when the license key or license file you supplied is not a StellarDiscover edition license. It also appears if the StellarOne server's primary license is not a StellarICS license, because StellarDiscover add-on licenses are supported only under a StellarICS primary license.	• Verify that the license key or license file you supplied is for the StellarDiscover edition. • Verify on the License page that the current license edition is StellarICS. • Contact your sales representative if you need a StellarICS license or a StellarDiscover add-on license.
Activation Code detected as the currently used license format. This trial license can only be used once. Please contact your sales representative for a trial License Key or upgrading to a full license.	Based on the terms and conditions, the trial Activation Code can only be used once. An example of the Activation Code: TE-24RF-Q9UN9-S9QQN-XXX XX-XXXXX-XXXXX See Comparison of License Formats (on page 69) for more details.	To continue using the product, contact your sales representative for a trial License Key or upgrading to a full license.
License Key detected as the currently used license format. Please contact your sales representative for extending the license.	Based on the terms and conditions, the trial License Key can be extended under certain circumstances.	To continue using the product, contact your sales representative for extending the trial license, upgrading to a full license, or renewing the full license.

Table 16. Licensing Error Messages and Suggested Actions (continued)

Error Message	Description	Possible Actions
	You can also choose to upgrade to a full license for a minimum period of one-year protection coverage. If you're using a full license, consider renewing the license. An example of the License Key: FIJN-HPYB-XXXX-XXXX	
Unable to connect to the license server. Proxy authentication is required.	This message signifies unspecified or invalid proxy authentication causes this issue that StellarOne cannot connect to the license server for license activation or renewal.	Check your proxy settings.
Unable to connect to the license server. The certificate was signed by an unknown authority.	This message signifies a self-signed certificate is used in a scenario that requires a trusted CA-signed certificate.	Import customer trusted certificate via vShell command. See <i>StellarOne Administrator Guide</i> for more details.
StellarOne has no Internet connection. You may try it again or choose to activate the license by importing the license file.	Due to unidentified network issues, StellarOne cannot connect to the license server to complete the license activation.	• Check your network settings and then try to activate or renew license again. • You can also use the License File to activate or renew license in the offline environment. The License Key is required for downloading the License File. See Getting the License File (on page 69) for more information.
StellarOne has no Internet connection. You may try it again or click New License to get and import the updated license file.	Due to unidentified network issues, StellarOne cannot connect to the license server to complete the license renewal.	
Unable to renew license. Make sure your network connection between StellarOne and license server is active or check your proxy settings and then try again.	Due to unidentified network issues, StellarOne cannot connect to the license server. Check your network settings and try again later.	Check your network settings and then try to renew license again.

Table 16. Licensing Error Messages and Suggested Actions (continued)

Error Message	Description	Possible Actions
	The Activation Code cannot be used to renew license offline.	

Collect Debug Log

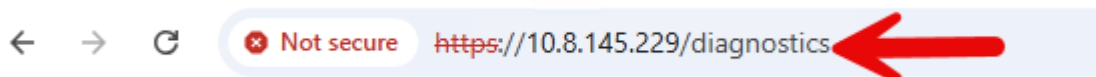
Before you contact TXOne Networks via the methods introduced in the following sections, you may need to obtain a debug log for the customer support agents to provide a solution. Follow the instructions below to obtain the Debug Log.



Note:

Only the account with the **Admin** role can collect the debug log.

1. Go to your StellarOne web console on a browser, in the address line, change the address after the server address to "**diagnostics**". Press **Enter**.



2. Set the **Debug Log Level** to **Verbose**, and then click **Save** to save the setting.
3. Perform the steps that originally led to the issue so that the debug logs can capture all necessary details.
4. Click the **Collect Debug Log** button.
5. Set the **Debug Log Level** back to **Warn**, and then click **Save** to save the setting.

Using the Support Portal

The TXOne Networks Support Portal is a 24x7 online resource that contains the most up-to-date information about both common and unusual problems.

1. Go to <https://help.txone.com/>.
2. Click the appropriate button to search for solutions.
3. Use the **Search** box to search for available solutions.
4. If no solution is found, click **Live Chat** or **VoIP** service to submit a support case online.

A TXOne Networks support engineer investigates the case and responds in 24 hours or less.



Important:

If the StellarOne virtual machine has been encrypted, ensure that you decrypt it for problem analysis.

Threat Encyclopedia

Most malware today consists of blended threats, which combine two or more technologies, to bypass computer security protocols. TXOne Networks combats this complex malware with products that create a custom defense strategy. The Threat Encyclopedia provides a comprehensive list of names and symptoms for various blended threats, including known malware, spam, malicious URLs, and known vulnerabilities.

Go to <https://www.encyclopedia.txone.com/> to learn more about:

- Malware and malicious mobile code currently active or "in the wild"
- Correlated threat information pages to form a complete web attack story
- Internet threat advisories about targeted attacks and security threats
- Web attack and online trend information
- Weekly malware reports

Contacting TXOne Networks

TXOne Networks representatives are available by phone or chat/VoIP services:

Table 17. TXOne Networks Contact Information

U.S.	+1 (346) 586-7975
Netherlands	+31 402-310-122
Taiwan	+886 (2) 7727-5120
Chat/VoIP services	https://help.txone.com/
Website	https://www.txone.com/contact/

- TXOne Networks product documentation:

<https://my.txone.com/>

Speeding Up the Support Call

To improve problem resolution, have the following information available:

- Steps to reproduce the problem
- Appliance or network information
- Computer brand, model, and any additional connected hardware or devices
- Amount of memory and free hard disk space
- Operating system and service pack version
- Version of the installed agent
- Product serial number and license file, or license key
- Detailed description of the environment where the agent is installed
- Exact text of any error message received

Other Resources

In addition to solutions and support, there are many other helpful resources available online to stay up to date, learn about innovations, and be aware of the latest security trends.

Download Center

From time to time, TXOne Networks may release a patch for a reported known issue or an upgrade that applies to a specific product or service. To find out whether any patches are available, go to:

<https://my.txone.com/>

If a patch has not been applied (patches are dated), open the Readme file to determine whether it is relevant to your environment. The Readme file also contains installation instructions.

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